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Laminations for transformers and inductors –

Part 1: Mechanical and electrical characteristics



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Laminations for transformers and inductors –

Part 1: Mechanical and electrical characteristics

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

LAMINATIONS FOR TRANSFORMERS AND INDUCTORS –

Part 1: Mechanical and electrical characteristics

FOREWORD

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International Standard IEC 60740-1 has been prepared by IEC technical committee 51: Magnetic components and ferrite materials.

IEC 60740-1 cancels and replaces IEC 60740 published in 1982 and its amendment 1 (1991). The main changes are listed below:

- a) ranges with lamination strips YS, YSUI 1, YSUI 2 and laminations YEE 2-...L added;
- b) range YEI 1 extended at the above end;
- c) ranges YED 2, YEF 2, YEL 2, YES 2, Type YM 1-5a and YM 1-7a cancelled;
- d) national designations cancelled;
- e) electrical characteristics for the laminations specified;
- f) mechanical characteristics for laminations added;
- g) holes added for lamination types YEI 1, YUI 1, YUI 2, YM 1;
- h) in Annex A, a conversion of the polarisation $\hat{J}$ and the field strength $\hat{H}$ in a specific total apparent power is defined. For the characteristics of the reactive power and the power loss, equations and constants are specified.

The text of this standard is based on the following documents:

FDIS	Report on voting
51/823/FDIS	51/836/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

IEC 60740 consists of the following parts under the general title *Laminations for transformers and inductors*:

Part 1: Mechanical and electrical characteristics

Part 2: Specification for the minimum permeabilities of laminations made of soft magnetic metallic materials.

The committee has decided that the contents of this publication will remain unchanged until the maintenance result date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- reconfirmed;
- withdrawn;
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

LAMINATIONS FOR TRANSFORMERS AND INDUCTORS –

Part 1: Mechanical and electrical characteristics

1 Scope

This part of IEC 60740 specifies the characteristics of laminations. Their preferred use is cores for transformers and inductors. The laminations are made of sheets and strips of magnetic materials, specified in IEC 60404-8-4 and IEC 60404-8-7.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60050-221, *International Electrotechnical Vocabulary (IEV) – Chapter 221: Magnetic materials and components*

IEC 60404-1:2000, *Magnetic materials – Part 1: Classification*

IEC 60404-8-4:1998, *Magnetic materials – Part 8-4: Specifications for individual materials – Cold-rolled non-oriented electrical steel sheet and strip delivered in the fully-processed state*

IEC 60404-8-7:1998, *Magnetic materials – Part 8-7: Specifications for individual materials – Cold-rolled grain-oriented electrical steel sheet and strip delivered in the fully-processed state*

IEC 60404-11:1999, *Magnetic materials – Part 11: Method of test for the determination of surface insulation resistance of magnetic sheet and strip*

IEC 61021-1:1990, *Laminated core packages for transformers and inductors used in telecommunication and electronic equipment – Part 1: Dimensions*

IEC 61021-2:1995, *Laminated core packages for transformers and inductors used in telecommunication and electronic equipment – Part 2: Electrical characteristics for cores using YEE 2 laminations*

ISO 286-1:1988, *ISO system of limits and fits – Part 1: Bases of tolerances, deviations and fits*

3 Terms, definitions and symbols

For the purposes of this document, the definitions of IEC 60050-221 and the following apply.

3.1

lamination

produced from a magnetic alloy sheet, usually consisting of one piece or several joined pieces, forming one complete layer of a laminated core

3.2

lamination strip

produced from a magnetic alloy sheet, which can be composed to a layer of laminations or stacks for limbs or yokes

3.3

square stack

results, if the height of the package h_p is equal to the limb width d .

3.4

specific power loss

loss of the core in an alternating field with specified frequency and sinusoidal waveform, generating a specified flux density divided by the core mass:

$$p_{Fe} = \frac{P_{Fe}}{m_{Fe}} \quad (1)$$

where

p_{Fe} is the specific power loss, in W/kg;

P_{Fe} is the power loss, in W;

m_{Fe} is the core mass, in kg.

NOTE 1 In the power loss both the hysteresis loss and eddy current loss are included.

NOTE 2 This is valid for cores with and without an air gap in the magnetic path.

3.5

specific reactive power

reactive power of the core in an alternating field by specified frequency and sinusoidal, specified flux density divided by the core mass:

$$p_{BFe} = \frac{P_{BFe}}{m_{Fe}} \quad (2)$$

where

p_{BFe} is the specific reactive power, in VA/kg;

P_{BFe} is the reactive power, in VA;

m_{Fe} is the core mass, in kg.

3.6

specific reactive power of the air gap

in an alternating field and sinusoidal, specified flux density is the r.m.s. reactive power in the air gap, divided by the core mass:

$$p_{BL} = 0,25 \cdot \frac{l_L \cdot f \cdot \hat{B}^2}{l_{Fe} \cdot \rho} \quad (3)$$

where

p_{BL} is the specific reactive power of the air gap, in VA/kg;

$\hat{B}$ is the peak flux density, in T;

f is the frequency, in Hz;

ρ is the density of the core, in kg/dm³;

l_L is the length of the air gap, in μm ;

l_{Fe} is the path length, in cm.

3.7

specific total apparent power of cores without air gap

in the mean flux path, the specific total apparent power consists of the specific reactive power and the specific power loss of the core and is the product of sinusoidal voltage and r.m.s. current divided by the core mass:

$$p_S = \sqrt{p_{BFe}^2 + p_{Fe}^2} \quad (4)$$

$$p_S = \frac{U \cdot I}{m_{Fe}} = \frac{P_S}{m_{Fe}} \quad (5)$$

where

P_S is the total apparent power, in VA;

p_S is the specific total apparent power, in VA/kg;

p_{BFe} is the specific reactive power, in VA/kg;

p_{Fe} is the specific power loss, in W/kg;

U is the voltage, in V;

I is the r.m.s. current, in A;

m_{Fe} is the core mass, in kg.

3.8

specific total apparent power of cores with air gap

in the mean flux path, the specific total apparent power consists of the specific reactive power of the core, the specific reactive power of the air gap and the specific power loss:

$$P_S = \sqrt{(p_{BFe} + p_{BL})^2 + p_{Fe}^2} \quad (6)$$

where

- p_S is the specific total apparent power, in VA/kg;
 p_{BFe} is the specific reactive power of the core, in VA/kg;
 p_{BL} is the specific reactive power of the air gap, in VA/kg;
 p_{Fe} is the specific power loss, in W/kg.

3.9

magnetic path length

arithmetic mean of the longest and the shortest path length of the core neglecting radii.

NOTE For details of the calculation for each core, see 13.3 to 13.10.

3.10 Symbols

- A_{Fe} core cross-section, in mm²;
 $\hat{B}$ peak flux density, in T;
 $\hat{B}_n$ nominal peak flux density, in T;
 $\hat{B}_0$ peak flux density for the characteristic of the specific reactive power, in dT;
 C_1 core constant 1, in mm⁻¹;
 C_2 core constant 2, in mm⁻³;
 c_1, c_2 constants for the characteristic of the specific reactive power;
 c_3, c_4 constants for the characteristic of the specific power loss;
 d limb width, in mm;
 f frequency, in Hz;
 $\hat{H}$ peak field strength, in A/m;
 h_p stack height, in mm;
 I current, in A;
 I_1, I_2, I_3 currents in the phases of three-phase cores, in A;
 $\hat{J}$ peak polarisation, in T;
 l magnetic path length, in mm;
 l_{Fe} magnetic path length, in mm;
 l_{Fe1}, l_{Fe2} magnetic path lengths for three-phase cores, in mm;
 l_L path length of the air gap, in µm;
 l_q magnetic path length cross to the rolling direction, in mm;
 l_w magnetic path length in rolling direction, in mm;
 m_{Fe} core mass, in kg;
 N number of turns;
 N_1 number of turns of winding N_1 ;
 N_2 number of turns of winding N_2 ;
 P_1, P_2, P_3 power loss in the phases of three-phase cores, in W;
 P_{BFe} reactive power of the core, in VA;
 P_{BL} reactive power of the air gap, in VA;
 P_{Fe} power loss, in W;

P_m	test value of the power loss, in W;
P_S	total apparent power, in VA;
p_B	specific reactive power, in VA/kg;
p_{BFe}	specific reactive power of the core, in VA/kg;
p_{BL}	specific reactive power of the air gap, in VA/kg;
p_{Fe}	specific power loss; in W/kg;
p_{Fe0}	cardinal value for the specific power loss, in W/kg;
p_S	specific total apparent power, in VA/kg;
p_{S0}	cardinal value of the total apparent power, in VA/kg;
$p_{S, \sin}$	specific total apparent power at sinusoidal current; in VA/kg;
R_1	resistance of winding N_1 , in Ω ;
R_2	resistance of winding N_2 , in Ω ;
R_n	precision resistor, in Ω ;
R_V	resistance of the voltmeter, in Ω ;
R_W	resistance of the voltage path of the wattmeter, in Ω ;
U	voltage, in V;
U_1	voltage of the source, in V;
$U_{2 \text{ r.m.s}}$	voltage of the winding N_2 , in V;
$U_{2 \text{ avg}}$	voltage of the winding N_2 , in V;
$\hat{U}_n$	peak value of the voltage across R_n , in V;
V_{Fe}	core volume, in mm ³ ;
x	tolerance factor;
y	auxiliary value for the characteristics of the specific reactive power;
y_1, y_2	constants for the characteristics of the specific reactive power;
y_3	constant for the characteristics of the specific power loss;
η	stacking factor;
μ_0	magnetic constant, in H/cm ($4 \cdot \pi \cdot 10^{-9}$ H/cm);
μ_a	amplitude permeability;
μ_i	initial permeability;
π	in this standard = 3,1416;
ρ	the density of the core alloy, in kg/dm ³ .

4 Materials and lamination thicknesses

4.1 Materials

Laminations are made from one of the materials specified in Table 1.

Table 1 – Preferred magnetic steel and alloys used for laminations

Material	Approximate composition in addition to iron	Density kg/dm ³	Designation	
			IEC 60404-1	IEC 60404-8-4 IEC 60404-8-7
Non-oriented silicon steel	1 % to 3 % silicon	7,65	C 21	M 270-35A 5 M 330-35A 5 M 330-50A 5
		7,7		M 400-50A 5 M 530-50A 5
		7,8		M 800-50A 5
Grain-oriented silicon steel	3,2 % silicon	7,65	C 22	M 165-35S 5
Nickel-iron alloys	72 % to 83 % nickel	8,7	E 1	—
	45 % to 50 % nickel (both oriented and non-oriented)	8,25	E 3	
	35 % to 40 % nickel	8,15	E 4	
Cobalt-iron alloys	47 % to 50 % cobalt (isotropic)		F 1	

4.2 Nominal lamination thickness

Lamination thickness shall be selected from one of the values specified in Table 2, according to the material in use.

Table 2 – Material and lamination thickness

Material designation	Thickness mm						
	0,5	0,38	0,35	0,3	0,2	0,1	0,05
C 21	0	-	0	-	-	-	-
C 22	-	-	0	-	-	-	-
E 1	-	x	x	0	0	0	x
E 3 non-oriented	-	x	x	0	0	0	x
E 3 grain-oriented	-	-	-	-	-	0	x
E 4	-	x	x	0	0	0	x
F 1	-	-	x	0	0	0	x
NOTE o: preferred value, x: usual, -: not usual.							

The allowed tolerance of the thickness of the laminations in Table 3 is corresponding to the appropriate value taken from IEC 60404-8-4 and IEC 60404-8-7.

Table 3 – Tolerance of the thickness of laminations

Material designation	Tolerances	
	mm	
	Thickness 0,5 mm	Thickness 0,35 mm
C 21	± 0,04	± 0,028
C 22	-	± 0,03

4.3 Minimum stacking factor

The minimum stacking factor η for usual materials and thickness shall be not lower than the values shown in Table 4. They are valid for uniform stacks with a minimum height of 10 mm subjected to a uniform pressure of 15 kN/m² perpendicular to the lamination.

Table 4 – Lamination thickness and stacking factor

Lamination thickness mm	Minimum stacking factor η		
	Non-oriented silicon steel C 21	Grain oriented silicon steel C 22	Nickel-iron and cobalt-iron alloys E, F
0,5	0,95	–	–
0,35	0,94	0,95	0,94
0,30	–	–	0,92
0,20	-	–	0,9
0,10	0,9	0,9	0,85
0,05	-	0,88	0,8

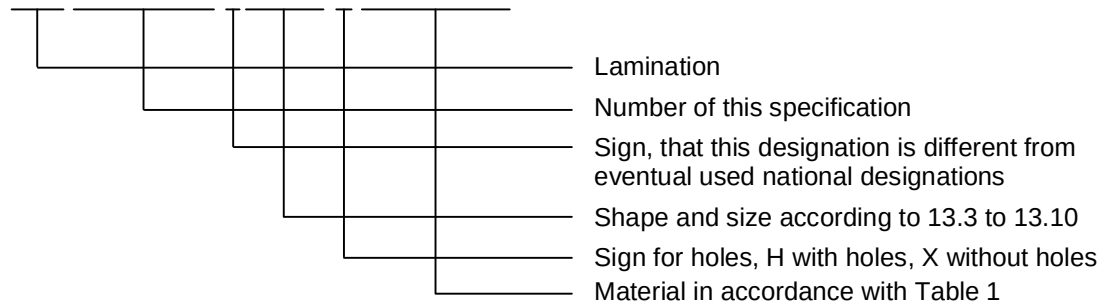
5 Designation of laminations and lamination strips

The designation contains all information needed to describe the laminations sufficiently and unmistakably.

5.1 Specified complete shapes

5.1.1 Lamination outline shapes

LAM IEC 60740-1 YUI 1-50 H M 530-50A 5



IEC 1195/05

IEC 1196/05

IEC 1197/05

IEC 1198/05

LAM IEC 60740-1 YS100 × 500 X M 165-35S 5

The diagram illustrates how a lamination designation code is structured. It shows a series of horizontal lines representing the components of the code, with vertical lines indicating their relative positions and groupings. The components are listed on the right side of the diagram:

- Lamination
- Number of this standard
- Sign, that this designation is different from eventual used national designations
- Lamination strips according to 13.2
- Width of the lamination strip in mm
- Length of the lamination strip in mm
- Sign for holes, H with holes, X without holes
- Material in accordance with Table 1

IEC 1199/05

The laminations shall be packed for transit in such a manner that no deterioration will occur. Unless otherwise specified, all parts designing a layer of lamination shall be packed in one packaging unit.

- lamination designation;
- quantity by mass or by number;
- the manufacturer's lot identification or date of manufacture;
- manufacturer.

- lamination: LAM IEC 60740-1 YEI 1-40 H M 330-35A 5
- quantity:
- date of manufacture:
- manufacturer:

7.1 General

7.2 Core constants

$$C_1 = \frac{I_{\text{Fe}}}{A_{\text{Fe}}} \quad (7)$$

$$C_2 = \frac{l_{\text{Fe}}}{A_{\text{Fe}}^2} \quad (8)$$

where

C_1 is the core constant 1, in mm^{-1} ;

C_2 is the core constant 2, in mm^{-3} ;

l_{Fe} is the magnetic path length, in mm, calculated as in 7.3;

A_{Fe} is the core cross-section, in mm^2 , taking into account the stacking factor η , see 7.4.

Further, the core volume, in mm^3 , and the core mass, in kg, are calculated as follows:

$$V_{\text{Fe}} = l_{\text{Fe}} \cdot A_{\text{Fe}} \quad (9)$$

$$m_{\text{Fe}} = V_{\text{Fe}} \cdot \rho \quad (10)$$

where ρ is the density of the alloy, in kg/dm^3 .

The effective parameters of laminated cores with square cross-section and a stacking factor of 0,95 are shown in Tables 17, 18, 19, 20, 21, 22, 23, 24, 25, 26 and 27. For the laminations in 13.3, 13.4.1 and 13.5.1 the equations for the core constants are given subsequent to the associated tables.

7.3 Magnetic path length

The magnetic path length l_{Fe} is calculated as the arithmetic mean of the longest and the shortest path length of the core neglecting radii.

NOTE For details on the calculation for each core, see 13.3, 13.4.1 and 13.5.1.

7.4 Core cross-section

The core cross-section is calculated as the product of (centre) limb width, stack height and stacking factor:

$$A_{\text{Fe}} = d \cdot h_p \cdot \eta \quad (11)$$

where

A_{Fe} is the core cross-section, in mm^2 ;

d is the limb width, in mm;

h_p is the stack height, in mm;

η is the stacking factor.

For cores with square stack ($h_p = d$) shown in Tables 17, 18, 19, 20, 21, 22, 23, 24, 25, 26 and 27 and with $\eta = 0,95$, the core cross-section is:

$$A_{\text{Fe}} = 0,95 \cdot d^2 \quad (12)$$

For other stack heights and stacking factors differing from Table 4, the values for A_{Fe} and C_1 shall be corrected by multiplying with the following factors:

The factor for A_{Fe} is:

$$\frac{h_p \cdot \eta}{d \cdot 0,95}$$

The factor for C_1 is:

$$\frac{d \cdot 0,95}{h_p \cdot \eta}$$

NOTE Alternatively, for a given core stack, the core cross-section may be determined as follows:

For YEI, YEx, YUI and YM:

$$A_{Fe} = \frac{m_{Fe}}{l_{Fe} \cdot \rho} \quad (13)$$

8 General conditions for electrical measurements

8.1 Atmospheric conditions

Unless otherwise stated, all tests shall be carried out at room temperature.

8.2 Test coils

The electrical tests in Clauses 15 and 16 are valid for the square stack. The stack height of the test coil and of the core may be changed, if the values are not changed. However, the test coils shall be suitable to the stack of laminations to be tested. The voltage winding N_2 shall be the inner winding and be as close as possible to the core, and its resistance R_2 shall be small compared to the resistance of its load.

The resistance in the supply circuit, including the coil resistance R_1 and the resistance of the current measuring device (current meter, current amplifier or precision resistor), shall be small, so that they do not result in any appreciable distortion of the voltage.

8.3 Test cores

The laminations shall physically be in accordance with the requirements of Clauses 4 and 13 and shall be selected at random from those which are fully processed for the end application. They shall be interleaved as a stack and shall consist of equal numbers of all necessary parts designing a complete layer of lamination.

NOTE Using grain-oriented and non-oriented silicon steel the laminations may be interleaved equilateral. Only in case of arbitration they shall be interleaved one by one throughout the stack.

Test cores made of lamination strips are a matter of agreement between customer and manufacturer.

8.4 Harmonic content of the voltage

In order to obtain comparable results, sinusoidal flux density shall be used for the measurements. During the test, the harmonic content shall not exceed 6 %. Where two voltmeters V_1 and V_2 in Figure 1a are connected in parallel to the winding N_2 , the first one is measuring the r.m.s. value and the second one is measuring the average value. The form factor shall be calculated from the ratio of the r.m.s. value to the average value. If this form factor is between 1,10 and 1,12 and at the same time the wave-shape as observed on the oscilloscope does not show obvious distortion, the harmonic content of the secondary voltage across the winding N_2 and hence of the flux density will be sufficiently small.

9 Measurements at high field strength

9.1 General

One of the circuits of Figure 1a, Figure 1b or Figure 2 shall be used. The test conditions shall be in accordance with the values specified in Table 5. Test certificates for lots of delivery are a matter of agreement between customer and manufacturer.

Table 5 – Test conditions for the materials C 21, C 22 and F 1 for power loss and total apparent power

Material designation	Thickness mm	Frequency Hz	Peak flux density T
Non-oriented silicon steel C 21	0,35 and 0,5	50 or 60	1,5
	0,1	400	1,0
Grain-oriented silicon steel C 22	0,35	50 or 60	1,7
	0,1 and 0,15	400	1,0
Cobalt-iron alloys F 1	0,2 and 0,35	50 or 60	2,0
	0,1	400	1,7

For the following equations the following symbols are used:

U_2	is the voltage on winding N_2 , in V;
f	is the frequency, in Hz;
$\hat{B}$	is the peak flux density, in T;
A_{Fe}	is the core cross-section, in mm ² (see 7.4);
I	is the current, in A;
I_1, I_2, I_3	currents in the phases of three-phase cores, in A;
m_{Fe}	is the core mass, in kg;
N_1	is the number of turns of the winding N_1 ;
N_2	is the number of turns of the winding N_2 ;
P_{Fe}	is the power loss, in W;
P_m	is the test value of power loss, in W;
P_S	is the total apparent power, in VA;
P_1, P_2, P_3	is the power loss in the phases of three-phase cores, in W;
R_2	is the resistance of the winding N_2 , in Ω ;
R_V	is the resistance of the voltmeter V_2 , in Ω ;
R_W	is the resistance of the voltage winding of the wattmeter, in Ω .

9.2 Measurement of power loss

9.2.1 Power loss for single-phase cores

Where the circuit of Figure 1a is used, both switches are closed for the determination of the form factor (see 8.4). Switch S_1 is then opened and the source voltage U_1 is adjusted so that the reading of the secondary voltage U_2 on the average sensing voltmeter V_2 has the value calculated from the following equation:

$$U_2 = \frac{2 \cdot \pi \cdot f}{\sqrt{2}} \cdot A_{Fe} \cdot \hat{B} \cdot N_2 \cdot \frac{1}{1 + R_2 \cdot \left(\frac{1}{R_V} + \frac{1}{R_W} \right)} \quad (14)$$

With switch S_1 open and switch S_2 closed, the wattmeter is read. Neglecting the copper losses in winding N_1 and in the ammeter, the power loss P_{Fe} are calculated with the following equation from the measured power loss P_m :

$$P_{Fe} = \left[P_m \cdot \frac{N_1}{N_2} - U_2^2 \cdot \left(\frac{1}{R_V} + \frac{1}{R_W} \right) \right] \cdot \left[1 + R_2 \cdot \left(\frac{1}{R_V} + \frac{1}{R_W} \right) \right] \quad (15)$$

NOTE 1 The result of the measurement is often expressed in terms of specific power loss $p_{Fe} = \frac{P_{Fe}}{m_{Fe}}$, in W/kg.

Where instruments incorporating linear preamplifiers (see Figure 1b) or similar devices with very high input impedance are used for voltage measurements, and with direct reading of the actual values of voltage and power, these equations are reduced to the expressions:

$$U_2 = \frac{2 \cdot \pi \cdot f}{\sqrt{2}} \cdot A_{Fe} \cdot \hat{B} \cdot N_2 \quad (16)$$

$$P_{Fe} = P_m \cdot \frac{N_1}{N_2} \quad (17)$$

NOTE 2 Where these impedances are not extremely high, but still much higher than R_2 , the following simplified equation may be used:

$$P_{Fe} = P_m \cdot \frac{N_1}{N_2} - U_2^2 \cdot \left(\frac{1}{R_V} + \frac{1}{R_W} \right) \quad (18)$$

Where no direct reading is possible, for example if external preamplifiers are used, the gains of these amplifiers shall be taken into account.

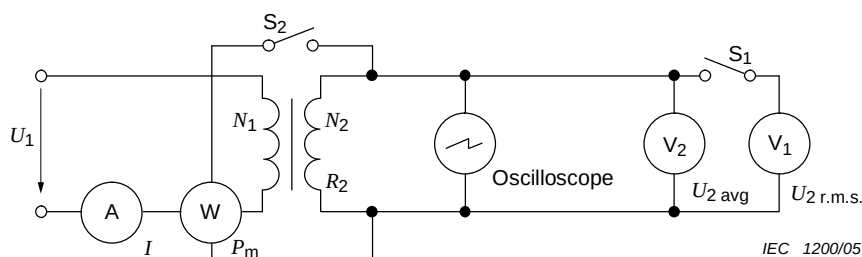


Figure 1a – Direct method

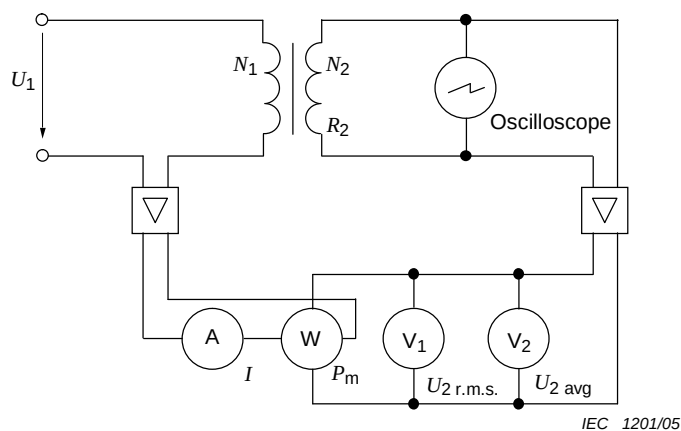


Figure 1b – Indirect method

Figure 1 – Measurements at high field strength, direct and indirect method, single-phase cores

9.2.2 Power loss for three-phase cores

The power loss in three-phase cores shall be measured with three watt meters using the circuit as shown in Figure 2. The induced secondary voltage shall be measured at the secondary winding of the middle limb. The total power loss are the sum of the three phases regarding the correct algebraic sign.

$$P_{Fe} = P_1 + P_2 + P_3 \quad (19)$$

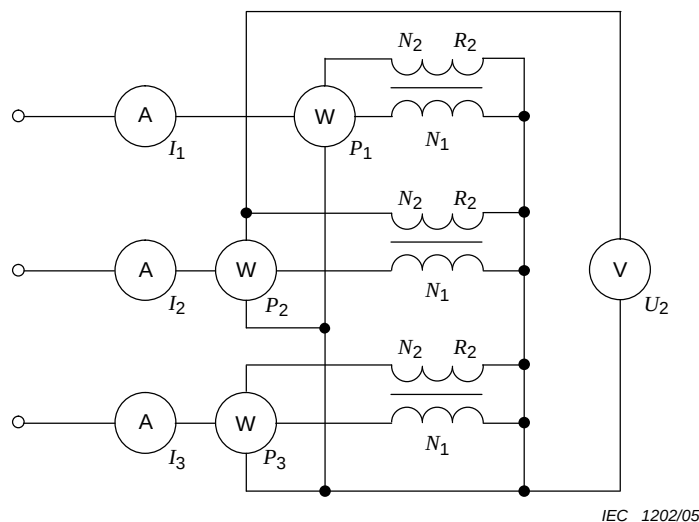


Figure 2 – Measurements at high field strength, direct method, three-phase cores

9.3 Measurement of total apparent power

9.3.1 Total apparent power for single-phase cores

Where the circuit of Figure 1a is used, both switches are closed in order to determine the form factor (see 8.4). Switch S_1 is then opened and the primary voltage adjusted as described in 9.2.1. With both switches S_1 and S_2 open, the primary current I and the secondary voltage U_2 are measured. The total apparent power is calculated using the following equation:

$$P_S = U_2 I \frac{N_1}{N_2} \cdot \frac{R_V + R_2}{R_V} \quad (20)$$

NOTE The result of the measurement is often expressed in terms of specific total apparent power $p_S = P_S / m_{Fe}$, in VA/kg (5).

Where instruments are used which incorporate linear preamplifiers (see Figure 1b) or similar devices with very high input impedance for voltage measurements, and with a direct reading of the actual values of voltage and power, this equation is reduced to the expression:

$$P_S = U_2 I \frac{N_1}{N_2} \quad (21)$$

Where no direct reading is possible, for example external preamplifiers are used, the gains of these amplifiers shall be taken into account.

9.3.2 Total apparent power for three-phase cores

The total apparent power for three-phase cores shall be measured as shown in Figure 2 and is calculated as the sum of the three currents multiplied with the voltage U_2 :

$$P_S = U_2 (I_1 + I_2 + I_3) \quad (22)$$

10 Measurements at low and medium field strength

10.1 General

The circuit of Figure 3 shall be used. The measuring conditions shall be taken from the values specified in Table 6 and Table 7. Test certificates for lots of delivery are a matter of agreement between customer and manufacturer.

Table 6 – Conditions for the measurement of amplitude permeability of cores using laminations of silicon steel

Thickness mm	Frequency Hz	Peak flux density T
0,05 up to 0,5	50 or 60	0,1 or 1,0

For the following equations the following symbols are used:

A_{Fe} is the core cross-section, in mm^2 (see 7.4);

$\hat{B}$ is the peak flux density, in T;

C_1 is the core constant 1, in cm^{-1} (see 7.2);

f is the frequency, in Hz;

$\hat{H}$ is the peak field strength, in A/m;

μ_0 is the magnetic constant, in H/cm ($4 \cdot \pi \cdot 10^{-9}$ H/cm);

μ_a is the amplitude permeability;

N_1 is the number of turns of the winding N_1 ;

N_2 is the number of turns of the winding N_2 ;

R_2 is the resistance of the winding N_2 , in Ω ;

R_n is the resistance of the precision resistor, in Ω ;

R_V is the resistance of the voltmeter V_2 , in Ω ;

U_2 is the voltage on winding N_2 , in V;

$\hat{U}_n$ is the peak value of the voltage across R_n , in V.

10.2 Measurement of amplitude permeability

The primary voltage is adjusted so that the reading of the r.m.s. voltage U_2 on the voltmeter V_2 meets the value calculated from the following equation:

$$U_2 = \frac{2 \cdot \pi \cdot f}{\sqrt{2}} \cdot A_{Fe} \cdot \hat{B} \cdot N_2 \cdot \frac{R_V}{R_V + R_2} \quad (23)$$

The peak value $\hat{I}$ of the primary current is measured by means of a peak voltmeter connected across a known precision resistor R_n in the primary circuit or by means of a probe. The amplitude permeability μ_a is calculated from the following equation:

$$\mu_a = \frac{1}{\mu_0} \cdot \frac{\hat{B}}{\hat{H}} = \frac{1}{\mu_0} \cdot \frac{\sqrt{2} \cdot U_2}{\hat{U}_n} \cdot \frac{C_1}{2 \cdot \pi \cdot f} \cdot \frac{1}{N_1 \cdot N_2} \cdot R_n \cdot \frac{R_V + R_2}{R_V}$$

(24)

- NOTE 1 For very small currents μ_a tends towards the initial permeability μ_i .
- NOTE 2 In some cases, it is convenient to use the complex permeability and its components.
- NOTE 3 Specification for the minimum permeability of laminations made of soft magnetic metallic materials, see IEC 60740-2.

Table 7 – Conditions for the measurement of amplitude permeability of cores using laminations of nickel-iron alloys

Material designation	Thickness mm	Preferred frequency Hz			Amplitude of field strength A/m
		50 or 60	300 or 400	1000	
E 1	0,05	x	x	x	0,2 or 0,4
	0,1	x	x	-	
	0,2	x	-	-	
	0,35	x	-	-	
E 3 (both oriented and non- oriented)	0,05	x	x	-	0,4
	0,1	x	x	-	
	0,2	x	-	-	
	0,35	x	-	-	
E 4	0,05	x	x	x	0,4 or 1,6
	0,1	x	x	-	
	0,2	x	-	-	
	0,35	x	-	-	

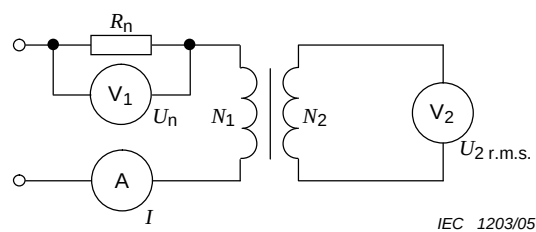


Figure 3 – Measurements at low and medium field strength

11 Preferred ranges of laminations

This clause describes the preferred shapes and ranges of limb width d . Table 8 shows a summary of these shapes and ranges with some general information.

Table 8 – Summary of preferred shapes

Shape	Clause	Limb width <i>d</i> mm	Description
YS	13.2	-	Lamination strips
YEI 1	13.3	10 up to 80	Range, allowing punching without scrap
YUI 1	13.4	10 up to 80	Range, allowing punching without scrap
YUI 2	13.5	10 up to 80	Range, allowing punching without scrap
YEx 2	13.6	2,4 up to 12	Range with small dimensions
YEE 2-..L	13.7	6 up to 12	Range with small dimensions, using two long E parts
YEx 3	13.8	10 up to 40	Range with small dimensions and with large window area
YEx 4	13.9	2,4 up to 12	Range with small dimensions and small mounting area
YM 1	13.10	5 up to 34	Range for laminations with low stray field

12 Dimensions and tolerances

12.1 Dimensions

The dimensions, or relation of dimensions, and tolerances of the preferred ranges of laminations are specified in the tables shown in 13.3 to 13.10. The tolerance on the lamination thickness is specified in Table 3.

12.2 Tolerances

The tolerances shall be in accordance with the requirements of ISO 286-1. The values corresponding to the reference designations at the head of each column of dimensions in the tables shown in 13.3 to 13.10 are taken from Table 9.

Table 9 – Tolerance according to ISO 286-1

Size range mm	ISO Tolerance code				
	IT 10 µm	IT 11 µm	IT 12 µm	IT 13 µm	IT 14 µm
up to 3	40	60	100	140	250
> 3 up to 6	48	75	120	180	300
> 6 up to 10	58	90	150	220	360
> 10 up to 18	70	110	180	270	430
> 18 up to 30	84	130	210	330	520
> 30 up to 50	100	160	250	390	620
> 50 up to 80	120	190	300	460	740
> 80 up to 120	140	220	350	540	870
> 120 up to 180	160	250	400	630	1000
> 180 up to 250	185	290	460	720	1150
> 250 up to 315	210	320	520	810	1300
> 315 up to 400	230	360	570	890	1400
> 400 up to 500	250	400	630	970	1550

NOTE The tolerance code indicates the value of the tolerance in relation to the nominal dimension.

For example, in Table 17, YEx 2-10:

size $a = 32$ mm with the tolerance IT 12 js = $a \pm 0,5 \times 250 \mu\text{m} = 32 \text{ mm} \pm 0,125 \text{ mm}$;
size $e = 22,4$ mm with the tolerance + IT 11 = $e + 130 \mu\text{m} = 22,4 \text{ mm} + 0,13 \text{ mm}$.

12.3 Symmetry tolerances

The symmetry tolerances of dimensions a and e with respect to the centre line shall not exceed half the tolerance specified for the width of the (centre) limb (dimension d).

13 Dimensions and effective parameters

13.1 General

In the following figures, the direction of rolling is indicated by a double arrow.

Subclauses 13.3 to 13.10 include the dimensions, the tolerances and the effective parameters, with their respective equations. The effective parameters are based on a stacking factor $\eta = 0,95$ and a stack height $h_p = d$. For the calculation, in the case of different stacking factors and heights, see 7.4.

13.2 Lamination strips Type YS

Lamination strips are specified as single strips with the nominal width d and the length a . For square lamination strips, the direction of rolling is to be indicated. Holes are a matter of agreement between customer and manufacturer.

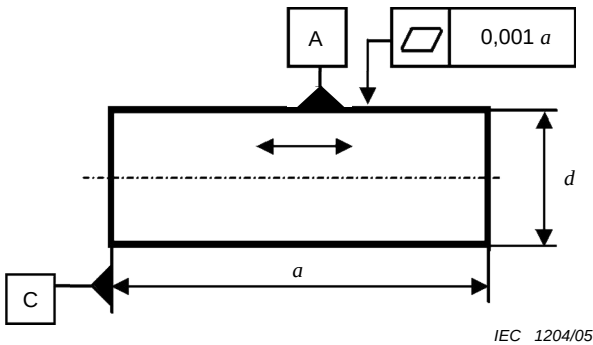


Figure 4 – Lamination strips Type YS, dimensions and tolerances

Table 10 – Tolerances for lamination strips

Length a mm	Tolerance mm	Width d mm	Tolerance mm
≤ 200	+0,25	≤ 150	+0,2
$> 200 \leq 400$	+0,3	$> 150 \leq 250$	+0,4
$> 400 \leq 800$	+0,35	> 250	+0,5
> 800	+0,5		

13.3 Laminations Type YEI 1

Laminations Type YEI 1, sizes from YEI 1-10 to YEI 1-80 are usual, where the numbers 10 to 80 are the nominal width of the centre limb.

Preferred dimensions d : 10, 13 (YEI 1-13 has a centre limb width 12,8 mm), 14, 16, 18, 20, 22, 25, 28, 32, 36, 40, 45, 50, 58, 64, 72 and 80 mm.

For the relation of the other dimensions of the lamination Type YEI 1 to the dimension d , see Table 11.

For the diameter for the holes h and for the tolerance factor x there is no fixed relation to dimension d , these dimensions are according to Table 12 in steps adjusted to dimension d . For representation of the dimensions, see Figure 5.

Laminations without holes are subject to an agreement between customer and manufacturer.

NOTE For laminations Type YEI 1, the coil formers IEC 61797-1, Type YEI 1, may be used.

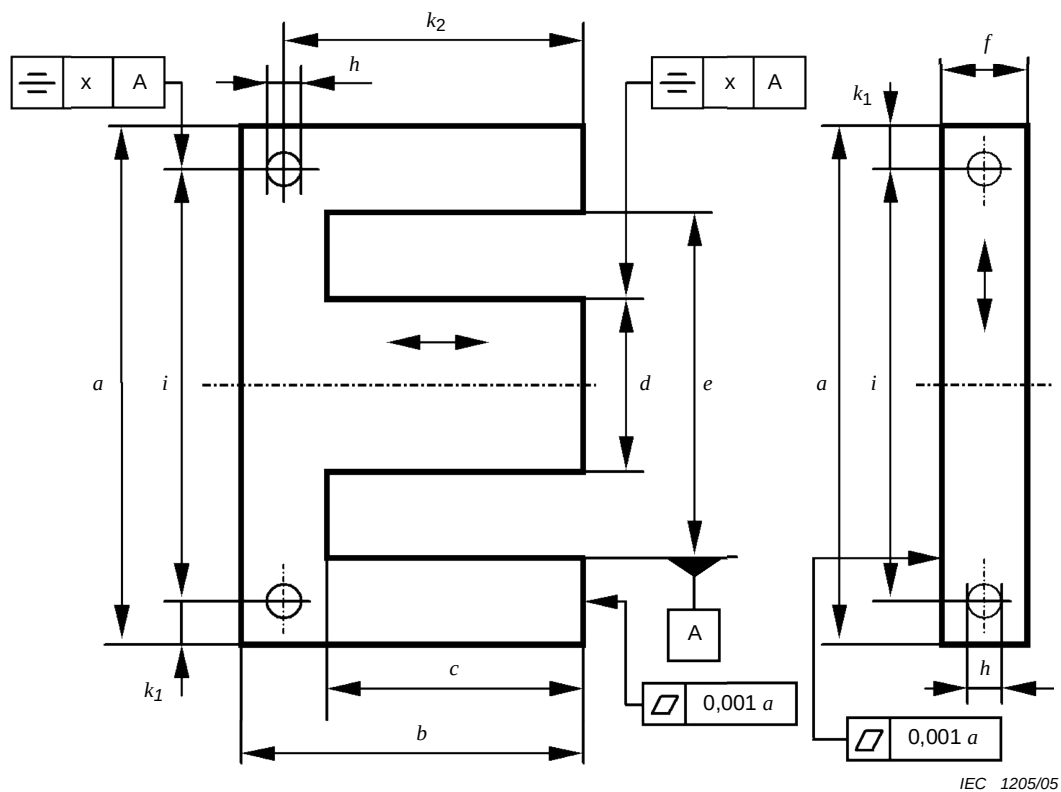


Figure 5 – Laminations Type YEI 1, dimensions

Table 11 – Laminations Type YEI 1, relation factor of dimensions and tolerances

Dimension	Relation factor	Tolerance code
<i>a</i>	3 <i>d</i>	IT 12 js
<i>b</i>	2 <i>d</i>	IT 12 js
<i>c</i>	1,5 <i>d</i>	IT 12 JS
<i>d</i>	1 <i>d</i>	IT 12 js
<i>e</i>	2 <i>d</i>	IT 12JS
<i>f</i>	0,5 <i>d</i>	IT 12 js
<i>i</i>	2,5 <i>d</i>	IT 12 JS
<i>k</i> ₁	0,25 <i>d</i>	-
<i>k</i> ₂	1,75 <i>d</i>	IT 12 JS

Table 12 – Laminations Type YEI 1, diameter of the holes and tolerance factor *x*

Dimension	Tolerance code	Lamination YEI 1-..., Dimension <i>d</i> mm							
		10 to 13	14 to 16	17 to 20	21 to 28	29 to 36	37 to 45	46 to 59	60 to 80
<i>h</i>	JS 14	-	3,5	3,5	4,5	5,5	6,6	8	10,5
<i>x</i>	-	0,1	0,1	0,2	0,2	0,2	0,4	0,4	0,5

The effective parameters are calculated as follows:

$$\text{Core cross-section} \quad A_{Fe} = d \cdot h_p \cdot \eta, \text{ in mm}^2 \quad (25)$$

$$\text{Magnetic path length} \quad l_{Fe} = b + c + f + (a + e - d)/2 = 6 \cdot d, \text{ in mm} \quad (26)$$

$$\text{Core volume} \quad V_{Fe} = A_{Fe} \cdot l_{Fe}, \text{ in mm}^3 \quad (27)$$

$$\text{Core constant} \quad C_1 = l_{Fe} / A_{Fe}, \text{ in mm}^{-1} \quad (28)$$

where

d is the limb width, in mm;

*h*_p is the stack height, in mm;

η is the stacking factor.

13.4 Laminations Type YUI 1

13.4.1 Lamination outline shapes Type YUI 1

Laminations Type YUI 1, sizes from YUI 1-10 to YUI 1-80 are usual, where the numbers 10 to 80 expresses the approximate width of the limb.

Preferred dimensions *d*: 10, 13, 16, 20, 25, 30, 34, 38, 44, 50, 56, 60, 70 and 80 mm.

For the relation of the other dimensions of the lamination Type YUI 1 to the dimension *d*, see Table 13.

For the diameter for the holes h and for the tolerance factor x there is no fixed relation to dimension d , these dimensions are according to Table 14 in steps adjusted to dimension d . Representation of the dimensions see Figure 6.

Laminations without holes are matter of agreement between customer and manufacturer.

NOTE For laminations Type YUI 1, the coil formers IEC 61797-1, Type YUI 1, may be used.

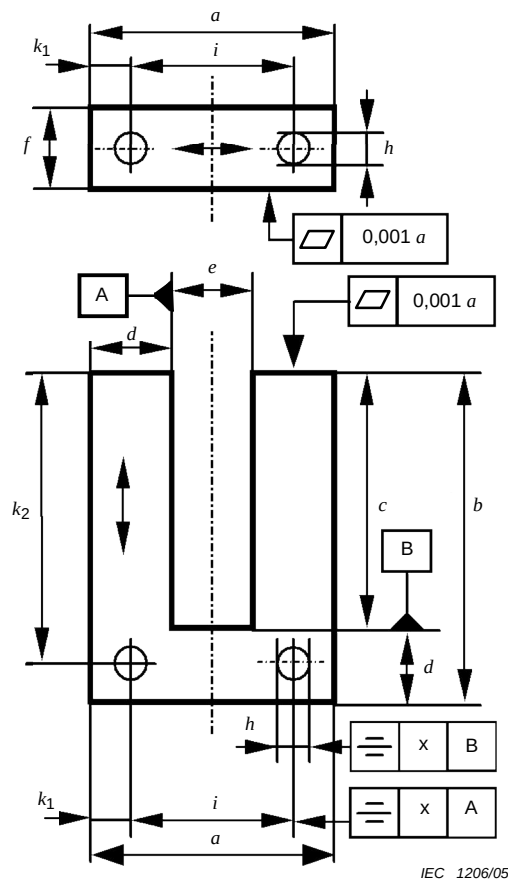


Figure 6 – Laminations Type YUI 1, dimensions

Table 13 – Laminations Type YUI 1, relation of dimensions and tolerances

Dimension	Relation factor	Tolerance code
a	$3 d$	IT 12 js
b	$4 d$	IT 12 js
c	$3 d$	IT 12 JS
d	$1 d$	-
e	$1 d$	IT 12 JS
f	$1 d$	IT 12 js
i	$2 d$	IT 12 JS
k_1	$0,5 d$	-
k_2	$3,5 d$	IT 12 JS

Table 14 – Laminations Type YUI 1, diameter of the holes and tolerance factor *x*

Dimen- sion	Tolerance code	Lamination YUI 1-... , dimension <i>d</i> mm							
		10 to14	15 to19	20 to 24	25 to 29	30 to 37	38 to 43	44 to 60	61 to 80
<i>h</i>	JS 14	3,5	4,5	4,5	5,5	7,8	11	11	15
<i>x</i>	-	0,1	0,1	0,2	0,2	0,2	0,2	0,4	0,4

The effective parameters are calculated as follows:

Core cross-section $A_{Fe} = d \cdot h_p \cdot \eta$, in mm² (29)

Magnetic path length $l_{Fe} = 2 \cdot b + 2 \cdot (a - d) = 12 \cdot d$, in mm (30)

Core volume $V_{Fe} = A_{Fe} \cdot l_{Fe}$, in mm³ (31)

Core constant $C_1 = l_{Fe} / A_{Fe}$, in mm⁻¹ (32)

where

d is the width of the limb, in mm;

h_p is the stack height, in mm;

η is the stacking factor.

13.4.2 Lamination strip shape Type YSUI 1

The lamination Type YUI 1 may also be constructed with lamination strips. For the layer plan of the separate strips, see Figure 7. The dimensions and tolerances of the strips are according to the dimensions and tolerances of the outline shapes. As a matter of agreement between customer and manufacturer the strip shapes may be provided with holes (sign H). The size and position of the holes shall be analogous in accordance with the holes in the outline shapes in Figure 6. Other holes are a matter for agreement between customer and manufacturer.

NOTE For lamination strip shapes Type YSUI 1, the coil formers IEC 61797-1, Type YUI 1, may be used.

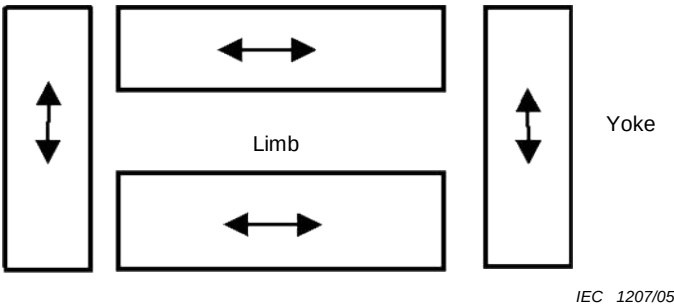


Figure 7 – Layer plan for the lamination strip shapes Type YSUI 1

13.5 Lamination Types YUI 2

13.5.1 Lamination outline shape Type YUI 2

Laminations Type YUI 2, sizes from YUI 2-10 to YUI 2-80 are usual, where the number 10 to 80 expresses the approximate width of the limb.

Preferred dimensions d : 10, 13, 16, 20, 25, 30, 34, 38, 44, 50, 56, 60, 70 and 80 mm.

For the relation of the other dimensions of the lamination Type YUI 2 to the dimension d , see Table 15.

For the diameter for the holes h and for the tolerance factor x , there is no fixed relation to dimension d , these dimensions according to Table 16 in steps are adjusted to dimension d . For representation of the dimensions, see Figure 8.

Laminations without holes are subject to an agreement between customer and manufacturer.

NOTE For laminations Type YUI 2 the coil formers IEC 61797-1, Type YUI 1, may be used.

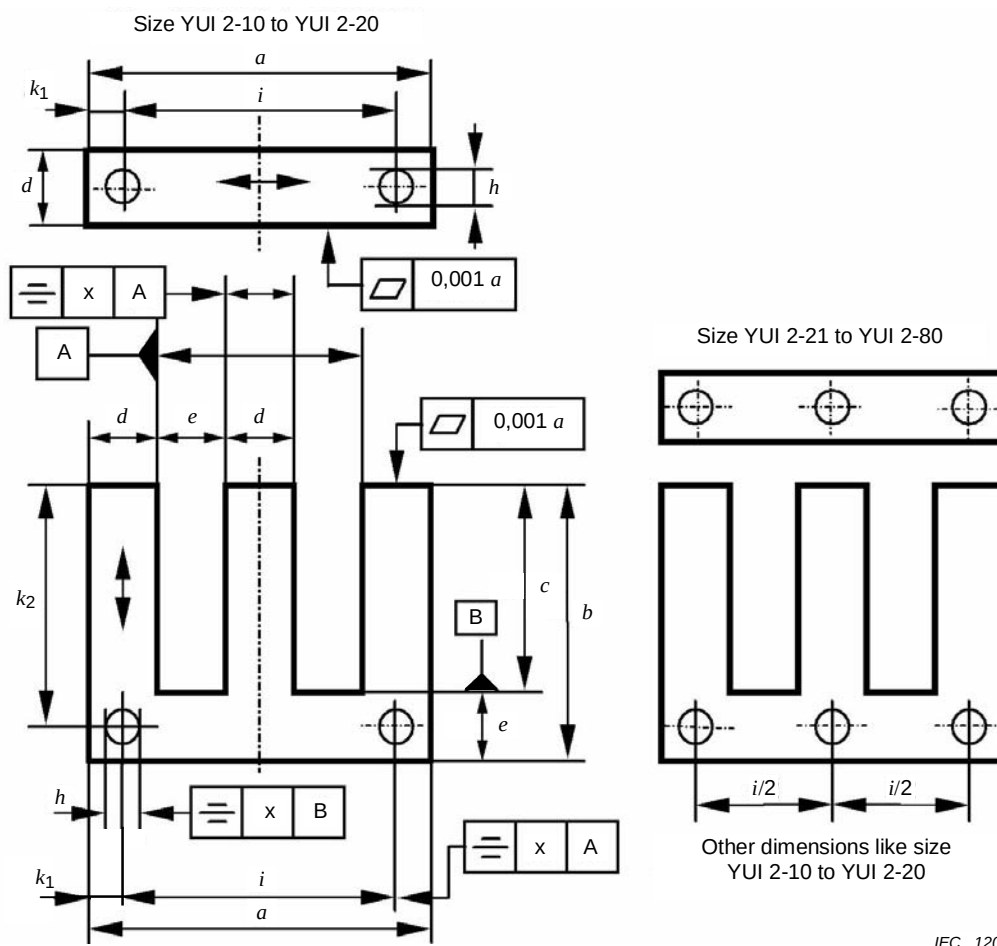


Figure 8 – Laminations Type YUI 2, dimensions

Table 15 – Laminations Type YUI 2, relation of dimensions and tolerances

Dimension	Relation factor	Tolerance code
a	$5 \cdot d$	IT 12 js
b	$4 \cdot d$	IT 12 js
c	$3 \cdot d$	IT 12 JS
d	$1 \cdot d$	IT 12 js
e	$1 \cdot d$	-
i	$4 \cdot d$	IT 12 JS
k_1	$0,5 \cdot d$	-
k_2	$3,5 \cdot d$	IT 12 JS

Table 16 – Laminations Type YUI 2, diameter of the holes and tolerance factor x

Dimension	Tolerance code	Lamination YUI 2-... , dimension d in mm						
		10 to 14	15 to 19	20 to 24	25 to 29	30 to 37	38 to 60	61 to 80
h	JS 14	3,5	4,5	4,5	5,5	7,8	11	15
x		0,1	0,1	0,2	0,2	0,2	0,4	0,4

The effective parameters are calculated as follows:

$$\text{Core cross-section} \quad A_{Fe} = d \cdot h_p \cdot \eta, \text{ in mm}^2; \quad (33)$$

$$\text{Magnetic path length} \quad l_{Fe1} = b + d, \text{ in mm}; \quad (34)$$

$$l_{Fe2} = b + 3 \cdot d, \text{ in mm}; \quad (35)$$

$$l_{Fe} = l_{Fe1} + 2 \cdot l_{Fe2} = 19 \cdot d; \quad (36)$$

$$\text{Core volume} \quad V_{Fe} = A_{Fe} \cdot l_{Fe}, \text{ in mm}^3; \quad (37)$$

$$\text{Core constant} \quad C_1 = l_{Fe} / A_{Fe}, \text{ in mm}^{-1}. \quad (38)$$

where

d is the limb width, in mm;

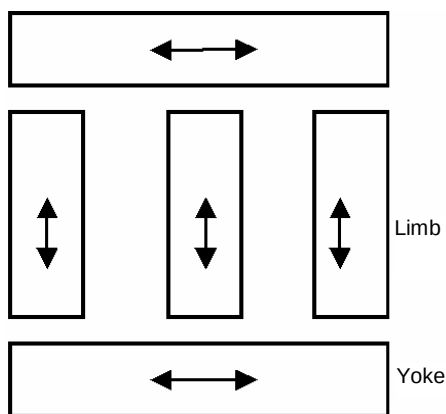
h_p is the stack height, in mm;

η is the stacking factor.

13.5.2 Lamination strip shape Type YSUI 2

The lamination Type YUI 2 also may be constructed with lamination strips. For the layer plan of the separate strips, see Figure 9. The dimensions and tolerances of the strips are according to the dimensions and tolerances of the outline shapes. As a matter of agreement between customer and manufacturer, the strip shapes may be provided with holes (sign H). The size and position of the holes shall be analogous in accordance with the holes in the outline shapes in Figure 8. Other holes are a matter for agreement between customer and manufacturer.

NOTE For lamination strip shapes Type YSUI 2, the coil formers IEC 61797-1, Type YUI 1, may be used.



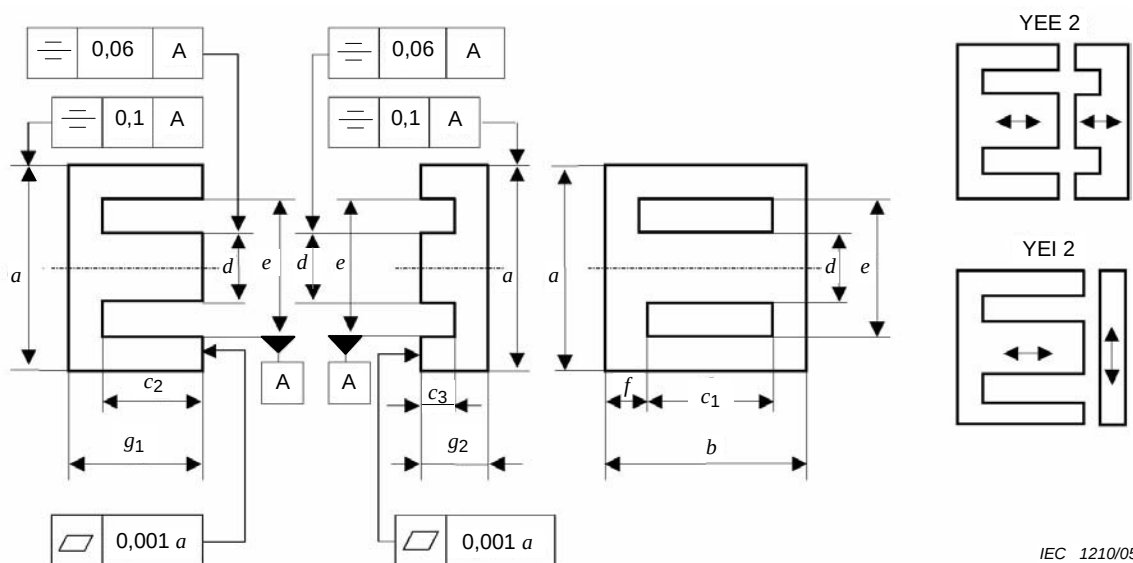
IEC 1209/05

Figure 9 – Layer plan for the lamination strip shapes Type YSUI 2

13.6 Lamination Types YEx 2, YEE 2 and YEI 2

YEx 2 designates YEE 2 or YEI 2. The corresponding type YEE 2-.. L is described in 13.7. For laminated core packages using YEE 2 laminations, see IEC 61021-1 and IEC 61021-2.

NOTE For laminations Type YEx 2, the coil formers IEC 61797-1, Type YEx 2, may be used.



IEC 1210/05

Figure 10 – Laminations Type YEx 2, dimensions

Table 17 – Laminations Type YEx 2, dimensions and tolerances

Designation	Reference letter and tolerance code (see Table 9)									
	<i>a</i>	<i>b</i>	<i>c</i> ₁	<i>c</i> ₂	<i>c</i> ₃	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i> ₁	<i>g</i> ₂
	IT 12 js	IT 13 js	+ IT 13 0	+ IT 12 0	+ IT 12 0	0 –IT 11	+ IT 11 0		IT 12 js	IT 12 js
YEx 2-2	8	8	5,6	4,3	1,3	2,4	5,6	1,2	5,5	2,5
YEx 2-3	10	10	7	5,5	1,5	3	7	1,5	7	3
YEx 2-4	12,6	12,6	8,8	6,7	2,1	3,8	8,8	1,9	8,6	4
YEx 2-5	16	16	11,2	8,6	2,6	4,8	11,2	2,4	11	5
YEx 2-6	20	20	14	11	3	6	14	3	14	6
YEx 2-8	25	25	17,4	13,2	4,2	7,6	17,4	3,8	17	8
YEx 2-10	32	32	22,4	17,2	5,2	9,6	22,4	4,8	22	10
YEx 2-12	40	40	28	22	6	12	28	6	28	12
Relation factor	3,33 · <i>d</i>	3,33 · <i>d</i>	2,33 · <i>d</i>			1 · <i>d</i>	2,33 · <i>d</i>	0,5 · <i>d</i>		

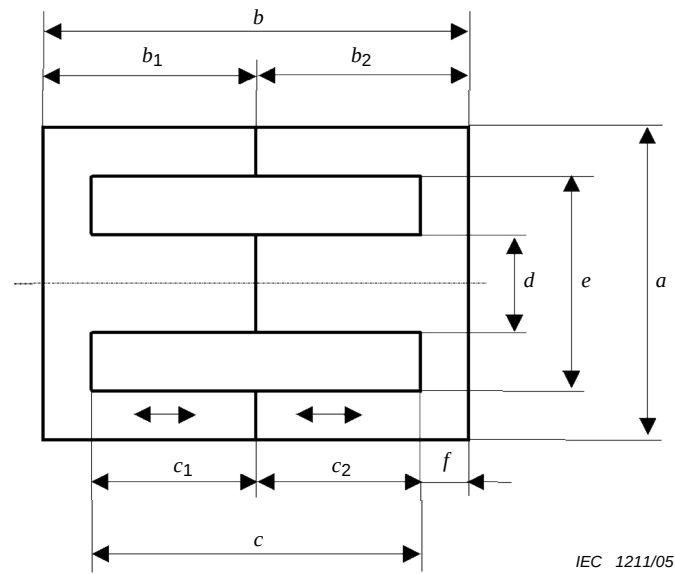
Table 18 – Laminations Type YEx 2, effective parameters

Designation	Effective parameters for $h_p = d$ and $\eta = 0,95$			
	Core cross-section	Magnetic path length	Core volume	Core constant
	A_{Fe} mm ²	l_{Fe} mm	V_{Fe} cm ³	C_1 cm ^{–1}
YEx 2-2	5,47	19,2	0,105	35,1
YEx 2-3	8,55	24,0	0,205	28,1
YEx 2-4	13,7	30,2	0,414	22,0
YEx 2-5	21,9	38,4	0,841	17,5
YEx 2-6	34,2	48,0	1,64	14,0
YEx 2-8	54,9	59,8	3,28	10,9
YEx 2-10	87,6	76,8	6,73	8,77
YEx 2-12	136,8	96,0	13,1	7,02

$$\text{Magnetic path length} \quad l_{Fe} = b + c_1 + (a + e - d) / 2 \quad (39)$$

13.7 Lamination Types YEE 2-..L

NOTE For laminations Type YEE 2-..L, the coil formers IEC 61797-1, Type YEE 2-..L, may be used.



$$b_1 = b_2 \quad c_1 = c_2$$

Figure 11 – Laminations Type YEE 2-...L, using two long E parts, dimensions

Table 19 – Laminations Type YEE 2-...L, using two long E parts, dimensions and tolerances

Designation	Reference letter and tolerance code (see Table 9)					
	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>
	IT 13 js	IT 14 js	IT 14 JS	0 -IT 11	+IT 11 0	
YEE 2-6 L	20	28	22	6	14	3
YEE 2-8 L	25	34	26,4	7,6	17,4	3,8
YEE 2-10 L	32	44	34,4	9,6	22,4	4,8
YEE 2-12 L	40	56	44	12	28	6
Relation factor	3,33 <i>d</i>	4,67 <i>d</i>	3,67 · <i>d</i>	1 · <i>d</i>	2,33 · <i>d</i>	0,5 · <i>d</i>

Table 20 – Laminations Type YEE 2-..L,
using two long E parts effective parameters

Designation	Effective parameters for $h_p = d$ and $\eta = 0,95$			
	Core cross-section	Magnetic path length	Core volume	Core constant
	A_{Fe} mm ²	l_{Fe} mm	V_{Fe} cm ³	C_1 cm ⁻¹
YEE 2-6 L	34,2	64	2,19	18,7
YEE 2-8 L	54,9	77,8	4,27	14,2
YEE 2-10 L	87,6	100,8	8,83	11,5
YEE 2-12 L	136,8	128	17,51	9,36

Magnetic path length $l_{Fe} = b + c + (a + e - d) / 2$ (40)

13.8 Lamination Types YEx 3, YEE 3, YEF 3, YEI 3 and YEL 3

YEx 3 designates YEE 3, YEF 3, YEI 3 or YEL 3.

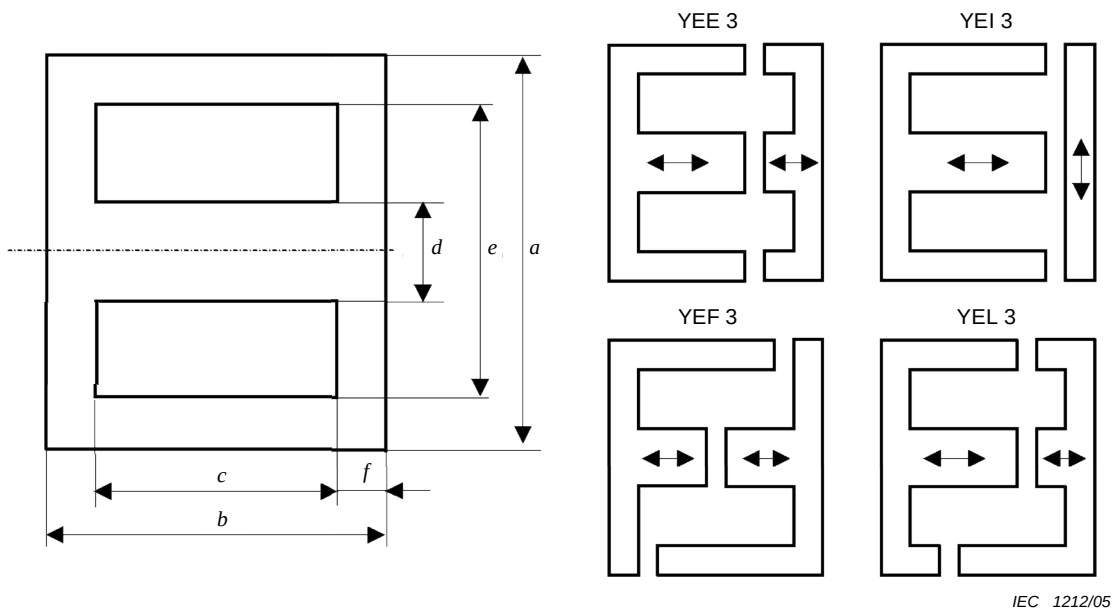


Figure 12 – Laminations Type YEx 3, dimensions

Table 21 – Laminations Type YEx 3, dimensions and tolerances

Designation	Reference letter and tolerance code (see Table 9)					
	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>
	± IT 12	± IT 12	+IT 12 0	0 -IT 12	+IT 12 0	
YEx 3-10	40	35	25	10	30	5
YEx 3-12	48	42	30	12	36	6
YEx 3-16	64	56	40	16	48	8
YEx 3-20	80	70	50	20	60	10
YEx 3-25	100	87,5	62,5	25	75	12,5
YEx 3-32	128	112	80	32	96	16
YEx 3-40	160	140	100	40	120	20
Relation factor	4 · <i>d</i>	3,5 · <i>d</i>	2,5 · <i>d</i>	1 · <i>d</i>	3 · <i>d</i>	0,5 · <i>d</i>

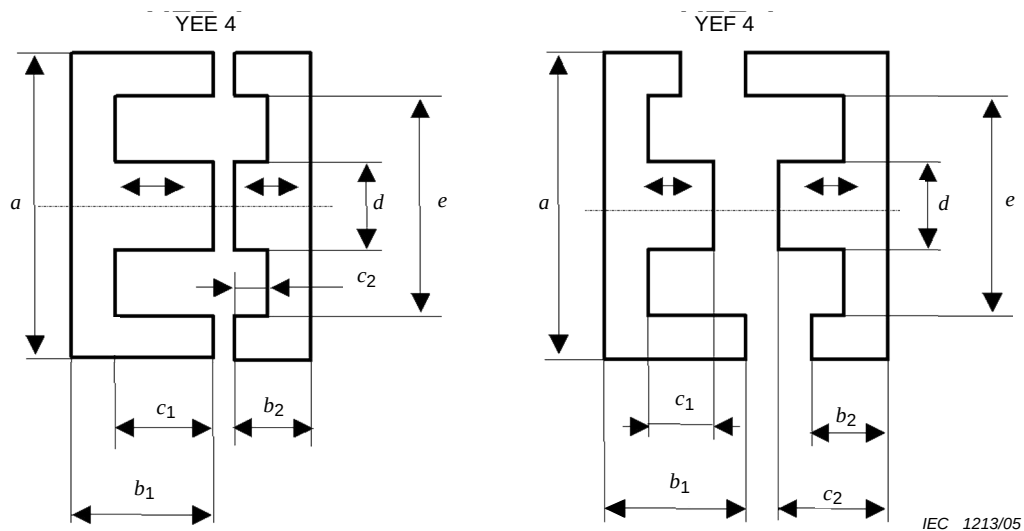
Table 22 – Laminations Type YEx 3, effective parameters

Designation	Effective parameters for $h_p = d$ and $\eta = 0,95$			
	Core cross-section A_{Fe} mm^2	Magnetic path length l_{Fe} mm	Core volume V_{Fe} cm^3	Core constant C_1 cm^{-1}
YEx 3-10	95	90	8,55	9,47
YEx 3-12	136,8	108	14,8	7,89
YEx 3-16	243,2	144	35,0	5,92
YEx 3-20	380	180	68,4	4,74
YEx 3-25	594	225	134	3,79
YEx 3-32	973	288	280	2,96
YEx 3-40	1 520	360	547	2,37

Magnetic path length $l_{Fe} = b + c + (a + e - d) / 2 = 9 \cdot d$ (41)

13.9 Lamination Types YEx 4, YEE 4 and YEF 4

YEx 4 designates YEE 4 or YEF 4.



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Figure 13 – Laminations Type YEx 4, dimensions

Table 23 – Laminations Type YEE 4, dimensions and tolerances

Designation	Reference letter and tolerance code (see Table 9)						
	<i>a</i>	<i>b</i> ₁	<i>b</i> ₂	<i>c</i> ₁	<i>c</i> ₂	<i>d</i>	<i>e</i>
	+IT 12 0	+IT 12 0	+IT 12 0	+IT 12 0	+IT 12 0	0 -IT 11	+IT 10 0
YEE 4-2	8,4	4	2	2,8	0,8	2,4	6
YEE 4-3	10,5	5	2,6	3,5	1,1	3	7,6
YEE 4-4	13,4	6,4	3,2	4,5	1,3	3,8	9,6
YEE 4-5	16,8	8	4	5,6	1,6	4,8	12
YEE 4-6	21	10	5	7	2	6	15
YEE 4-8	26,6	12,7	6,3	8,9	2,5	7,6	19
YEE 4-10	33,6	16	8	11,2	3,2	9,6	24
YEE 4-12	42	20	10	14	4	12	30
Relation factor	3,5 · <i>d</i>	1,67 · <i>d</i>	0,83 · <i>d</i>	1,17 · <i>d</i>	0,33 · <i>d</i>	1 · <i>d</i>	2,5 · <i>d</i>

Table 24 – Laminations Type YEF 4, dimensions and tolerances

Designation	Reference letter and tolerance code (see Table 9)						
	<i>a</i>	<i>b</i> ₁	<i>b</i> ₂	<i>c</i> ₁	<i>c</i> ₂	<i>d</i>	<i>e</i>
	+IT 12 0	0 -IT 12	0 -IT 12	+IT 12 0	+IT 12 0	0 -IT 11	+IT 10 0
YEF 4-2	8,4	4	2	1,8	3	2,4	6
YEF 4-3	10,5	5	2,6	2,3	3,8	3	7,6
YEF 4-4	13,4	6,4	3,2	2,9	4,8	3,8	9,6
YEF 4-5	16,8	8	4	3,6	6	4,8	12
YEF 4-6	21	10	5	4,5	7,5	6	15
YEF 4-8	26,6	12,7	6,3	5,7	9,5	7,6	19
YEF 4-10	33,6	16	8	7,2	12	9,6	24
YEF 4-12	42	20	10	9	15	12	30
Relation factor	3,5 · <i>d</i>	1,67 · <i>d</i>	0,83 · <i>d</i>	0,75 · <i>d</i>	1,25 · <i>d</i>	1 · <i>d</i>	2,5 · <i>d</i>

Table 25 – Laminations Type YEx 4, effective parameters

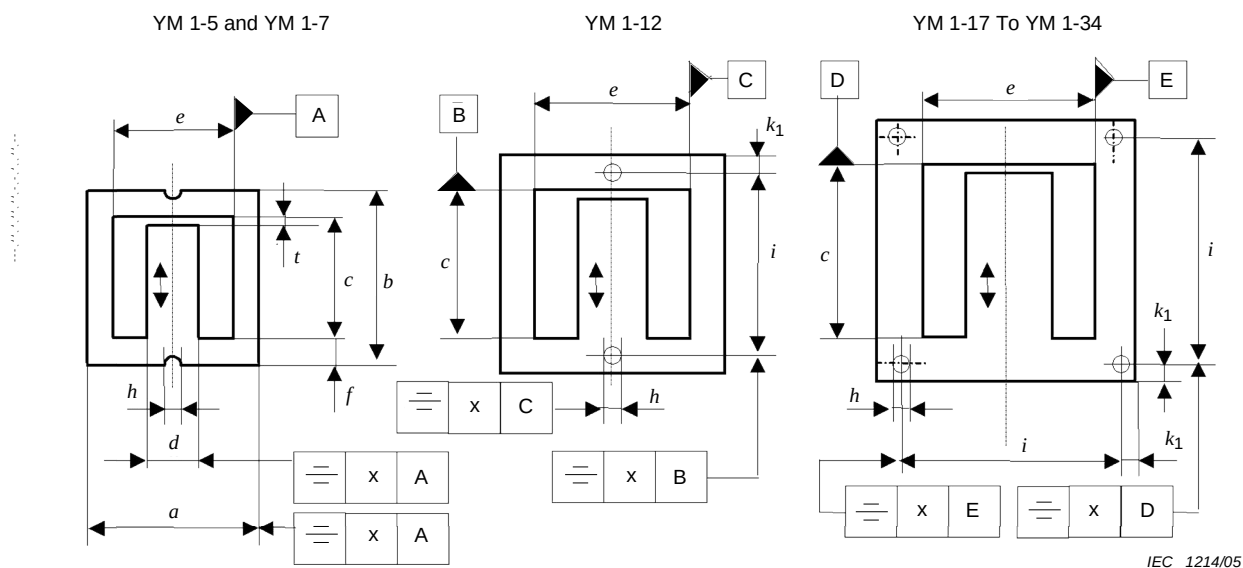
Designation	Effective parameters for $h_p = d$ and $\eta = 0,95$			
	Core cross-section A_{Fe} mm ²	Magnetic path length l_{Fe} mm	Core volume V_{Fe} cm ³	Core constant C_1 cm ⁻¹
YEx 4-2	5,47	15,2	0,0832	27,8
YEx 4-3	8,55	19,75	0,169	23,1
YEx 4-4	13,72	25	0,343	18,2
YEx 4-5	21,89	31,2	0,683	14,3
YEx 4-6	34,2	39	1,33	11,4
YEx 4-8	54,87	49,4	2,71	9,0
YEx 4-10	87,55	62,4	5,46	7,13
YEx 4-12	136,8	78	10,7	5,70

$$\text{Magnetic path length for type YEE: } l_{Fe} = b_1 + b_2 + c_1 + c_2 + (a + e - d) / 2 = 6,5 \cdot d \quad (42)$$

$$\text{Magnetic path length for type YEF: } l_{Fe} = 2 \cdot c_1 + 2 \cdot c_2 + (a + e - d) / 2 = 6,5 \cdot d \quad (43)$$

13.10 Laminations Type YM 1

NOTE For laminations Type YM 1, the coil formers IEC 61797-1, Type YM 1, may be used.



For other dimensions see YM 1-5

Figure 14 – Laminations Type YM 1, dimensions

Table 26 – Laminations Type YM 1, dimensions and tolerances

Designation	Reference letter and tolerance code (see Table 9)										
	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>h</i>	<i>i</i>	<i>k</i> ₁	<i>t</i> ^{a)}	<i>x</i>
	IT 12 js	IT 12 js	IT 12 JS	IT 12 js	IT 12 JS		JS 14	IT 12 JS		± IT 11	
YM 1-5	20	20	13	5	13	3,5	2,8				0,1
YM 1-7	30	30	20	7	20	5	3,0				0,1
YM 1-12	42	42	30	12	30	6	3,5	36	3	0,5	0,1
YM 1-17	55	55	38	17	38	8,5	3,5	47	4	0,5	0,2
YM 1-20	65	65	45	20	45	10	4,5	56	4,5	0,5	0,2
YM 1-23	74	74	51	23	51	11,5	4,5	64	5	0,5	0,2
YM 1-29	85	85	56	29	56	14,5	4,5	75	5	0,5	0,2
YM 1-34	102	102	68	34	68	17	5,5	91	5,5	0,5	0,2
a) for <i>t</i> = 0 the tolerance is + 0,2 mm											

Table 27 – Laminations Type YM 1, effective parameters

Designation	Effective parameters , $h_p = d$ and $\eta = 0,95$			
	Core cross-section	Magnetic path length	Core volume	Core constant
	A_{Fe} mm ²	l_{Fe} mm	V_{Fe} cm ³	C_1 cm ⁻¹
YM 1-5	23,75	47	1,12	19,8
YM 1-7	46,55	71,5	3,33	15,4
YM 1-12	136,8	102	14	7,46
YM 1-17	274,55	131	36	4,77
YM 1-20	380	155	58,9	4,08
YM 1-23	502,55	176	88,4	3,5
YM 1-29	798,95	197	157	2,47
YM 1-34	1 098,2	238	261	2,17

Magnetic path length
$$l_{Fe} = b + c + (a + e - d) / 2 \quad (44)$$

14 Electrical characteristics for materials

For the electrical characteristics of the laminations a definition of specific total apparent power in VA/kg is required.

A conversion of the polarisation $\hat{J}$ and the field strength $\hat{H}$ from IEC 60404-8-4 and IEC 60404-8-7 into equivalent specific total apparent power is obtained with a process that uses guaranteed values for $\hat{J}$ and $\hat{H}$ along with a typical curve of the apparent power.

For each material in Table 1, a cardinal value p_{S0} , of total apparent power for the nominal flux density $\hat{B}$ is defined. See Annex A.

For the specific power loss from IEC 60404-8-4 and IEC 60404-8-7, cardinal values p_{Fe0} in W/kg are defined, which are higher than the limits of power loss for sheets and strips.

The selected cardinal values for the specific total apparent power p_{S0} and the specific power loss p_{Fe0} are in accordance with the series R 40.

**Table 28 – Materials from IEC 60404-8-4 and IEC 60404-8-7 and Table 1.
Characteristics for the electrical steel sheet and strips, and laminations
for the specific total apparent power p_{S0} and the specific power loss p_{Fe0}**

Material designation	Electrical sheet and strip					Laminations				
	IEC 60404-8-4 IEC 60404-8-7					IEC 60740-1				
	$\hat{H}$ A/m	$\hat{B}, \hat{J}$ T	p_S ^{a)} VA/kg	$\hat{B}$ T	p_{Fe} W/kg	$\hat{B}$ T	p_{S0} VA/kg	p_{Fe0} ^{b)} W/kg	Cores made from	
M 270-35-A 5	2 500	1,49	53	1,5	2,7	1,5	56	3	Contour outline shapes	
M 330-35-A 5					3,3			3,75		
M 330-50-A 5					3,3			3,75		
M 400-50-A 5					4			50		4,5
M 530-50-A 5					5,3			40		6
M 800-50-A 5					8			31,5		9
M 165-35-S 5	800	1,75	21,3	1,7	1,65	1,7	50	2,12	Strip shapes	
							56	2,8	Contour outline shapes	
a) The values for p_S are valid for the flux density $\hat{B}$, $\hat{J}$. Conversion from $\hat{H}$ and $\hat{J}$ to p_{S0} , see annex A.										
b) In the values p_{Fe0} at the non-oriented material C 21, the parts of anisotropy are included.										

Laminations made from grain-oriented silicon steel have essential higher total apparent power. The cardinal value $p_{S0} = 56$ VA/kg at $\hat{B} = 1,7$ T for the laminations is chosen according to the practice.

15 Specific total apparent power for laminations

The cardinal values p_{S0} in Table 28 are valid for Epstein trails with a width of 30 mm.

The limits for the total apparent power p_S for lamination strip shapes and outline shapes are specified in Table 29. They are valid for the materials of Table 1, for all types and width of the limbs of the laminations.

At the laminations with limbs from 6 mm to 14 mm, the limits are valid without holes in the laminations. At limbs from 15 mm to > 60 mm the values are valid with the holes from 13.3 to 13.5 and 13.10. Other holes are a matter for agreement between customer and manufacturer.

The limits for p_S in small laminations with holes are about one step of R 40 higher than for those without holes. All values for the specific total apparent power p_S are in accordance with series R 40. The laminations may be interleaved equilateral, only in the case of arbitration they shall be interleaved one by one throughout the stack.

Table 29 – Specific total apparent power p_S in VA/kg for the materials from Table 1 for all types of laminations and limb width

Material designation	Type	$\hat{B}$ T	p_{S0} VA/kg	p_S in VA/kg			
				Lamination limb width			
				mm			
				6 to 14	15 to 29	30 to 60	> 60
				without holes	with holes		
M 270-35A 5, M 330-35A 5, M 330-50A 5	YS	1,5	56	-	-	60	56
	YEI 1, YUI 1	1,5	56	75	71	60	56
	YSUI 1	1,5	56	-	-	60	56
	YUI 2	1,5	56	75	75	63	60
	YSUI 2	1,5	56	-	-	63	60
	YM 1	1,5	56	80	80	67	-
M 400-50A 5	YS	1,5	50	-	-	53	50
	YEI 1, YUI 1	1,5	50	67	63	53	50
	YSUI 1	1,5	50	-	-	53	50
	YUI 2	1,5	50	67	67	56	53
	YSUI 2	1,5	50	-	-	56	53
	YM 1	1,5	50	71	71	60	-
M 530-50A 5	YS	1,5	40	-	-	42,5	40
	YEI 1, YUI 1	1,5	40	53	50	42,5	40
	YSUI 1	1,5	40	-	-	42,5	40
	YUI 2	1,5	40	53	53	45	42,5
	YSUI 2	1,5	40	-	-	45	42,5
	YM 1	1,5	40	56	56	47,5	-
M 800-50A 5	YS	1,5	31,5	-	-	33,5	31,5
	YEI 1, YUI 1	1,5	31,5	42,5	40	33,5	31,5
	YSUI 1	1,5	31,5	-	-	33,5	31,5
	YUI 2	1,5	31,5	42,5	42,5	35,5	33,5
	YSUI 2	1,5	31,5	-	-	35,5	33,5
	YM 1	1,5	31,5	45	45	37,5	-
M 165-35S 5	YS, YSUI 1	1,7	50	-	-	56	50
	YSUI 2	1,7	56	-	-	63	56
	YEI 1, YUI 1	1,7	56	85	85	67	56
	YUI 2	1,7	56	90	90	71	60
	YSUI 2	1,7	56	90	90	71	60
	YM 1	1,7	56	100	100	80	-

16 Specific power loss for laminations

The cardinal values p_{Fe0} in Table 28 are valid for Epstein trails with a width of 30 mm.

The limits for the power loss p_{Fe} for strips and laminations are in Table 30. They are valid for the materials from Table 1, for all types and width of the limb of the laminations.

At the laminations with limbs from 6 mm to 14 mm, the limits are valid without holes in the laminations. At limbs from 15 mm to > 60 mm, the values are valid with the holes from 13.3 to 13.5 and 13.10. Other holes are a matter of agreement between customer and manufacturer.

The limits for p_{Fe} in the small laminations with holes are about one step of R 40 higher than without holes. All values for p_{Fe} are in accordance with the series R 40. The laminations may be interleaved equilateral, only in the case of arbitration they shall be interleaved one by one throughout the stack.

Table 30 – Specific power loss p_{Fe} in W/kg for the materials from Table 1 for all types of laminations and limb width

Material designation	Type	$\hat{B}$ T	p_{Fe0} W/kg	p_{Fe} in W/kg			
				Lamination limb width in mm			
				6 to 14	15 to 29	30 to 60	> 60
				without holes	with holes		
M 270-35A 5	YS	1,5	3	-	-	3,15	3
	YEI 1, YUI 1	1,5	3	3,55	3,35	3	3
	YSUI 1	1,5	3	-	-	3	3
	YUI 2	1,5	3	3,75	3,55	3,15	3,15
	YSUI 2	1,5	3	-	-	3,15	3,15
	YM 1	1,5	3	3,75	3,55	3,15	-
M 330-35A 5, M 330-50A 5	YS	1,5	3,75	-	-	4	3,75
	YEI 1, YUI 1	1,5	3,75	4,5	4,25	3,75	3,75
	YSUI 1	1,5	3,75	-	-	3,75	3,75
	YUI 2	1,5	3,75	4,75	4,5	4	4
	YSUI 2	1,5	3,75	-	-	4	4
	YM 1	1,5	3,75	4,75	4,5	4	-
M 400-50A 5	YS	1,5	4,5	-	-	4,75	4,5
	YEI 1, YUI 1	1,5	4,5	5,3	5	4,5	4,5
	YSUI 1	1,5	4,5	-	-	4,5	4,5
	YUI 2	1,5	4,5	5,6	5,3	4,75	4,75
	YSUI 2	1,5	4,5	-	-	4,75	4,75
	YM 1	1,5	4,5	5,6	5,3	4,75	-
M 530-50A 5	YS	1,5	6	-	-	6,3	6
	YEI 1, YUI 1	1,5	6	7,1	6,7	6	6
	YSUI 1	1,5	6	-	-	6	6
	YUI 2	1,5	6	7,5	7,1	6,3	6,3
	YSUI 2	1,5	6	-	-	6,3	6,3
	YM 1	1,5	6	7,5	7,1	6,3	-
M 800-50A 5	YS	1,5	9	-	-	9,5	9
	YEI 1, YUI 1	1,5	9	10,6	10	9	9
	YSUI 1	1,5	9	-	-	9	9
	YUI 2	1,5	9	11,2	10,6	9,5	9,5
	YSUI 2	1,5	9	-	-	9,5	9,5
	YM 1	1,5	9	11,2	10,6	9,5	-
M 165-35S 5	YS, YSUI 1, YSUI 2	1,7	2,12	-	-	2,24	2,12
	YEI 1, YUI 1	1,7	2,8	3,15	3	2,8	2,8
	YUI 2, YM 1	1,7	2,8	3,35	3,15	3	3

17 Mechanical characteristics for laminations

17.1 General appearance and conditions

Laminations shall be clean and free from oil and rust. They shall be flat, and have no cracks and other irregularities. Heat treated laminations shall be free from contaminating deposits.

17.2 Cut and punch edge

For laminations made of materials from Table 1, the cut and punch edge shall not exceed + 0,05 mm on a length of 10 mm. Punctual edges are allowed up to 0,1 mm.

17.3 Camber

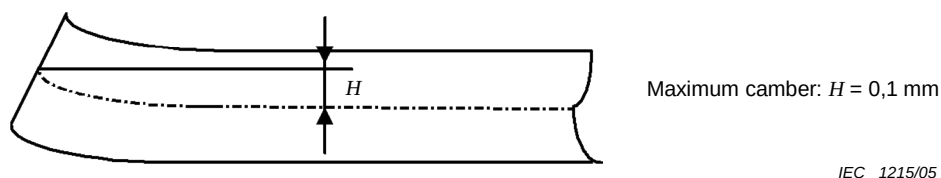


Figure 15 – Definition and limit of camber

17.4 Distortion

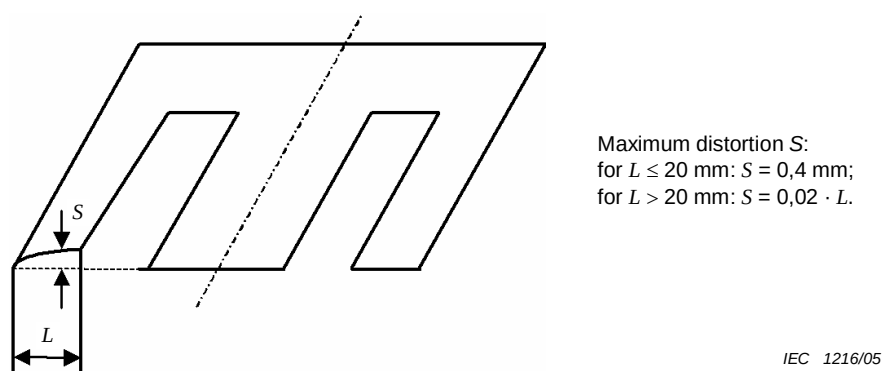


Figure 16 – Definition and limit of distortion

17.5 Bending

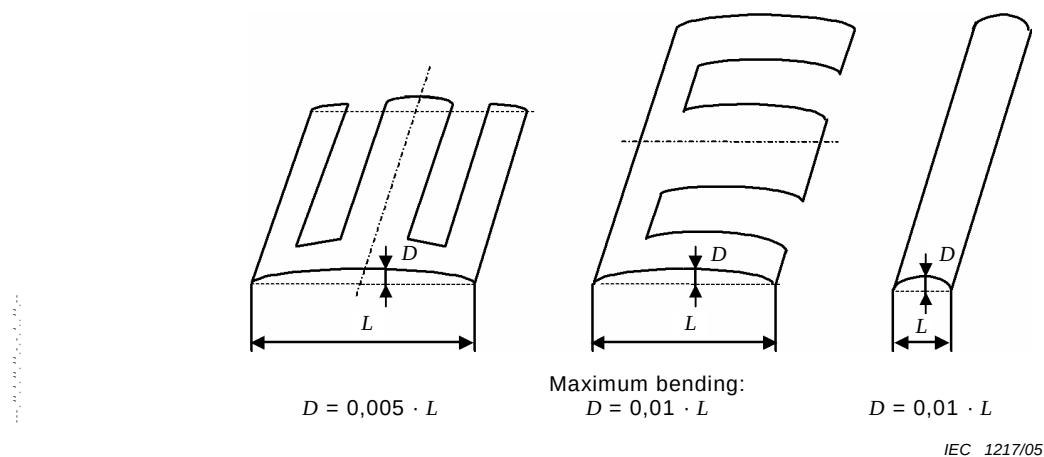


Figure 17 – Definition and limit of bending

17.6 Insulation

Laminations shall be isolated from each other.

For laminations made of material C 21 or C 22 the insulation shall consist of coating material made of inorganic substances, however a small part of organic substances may be included in order not to deteriorate the ability of welding according to 17.7.

Laminations made of the materials E 1, E 3, E 4 and F 1 shall be insulated with coating of fine-dispersed inorganic substances.

The insulation shall have an insulation resistance in according to Table 31. The measurement procedure shall be in accordance with IEC 60404-11.

Table 31 – Insulation resistance of coated surface of laminations made from material according to Table 1

Material	Electrical resistance $\Omega \cdot \text{cm}^2$
Laminations from non-oriented steel C 21	≥ 5
Laminations from grain-oriented steel C 22	≥ 5
Laminations from E 1, E 3, E 4 and F 1	Not specified

NOTE At the present time, there are not standardized laminations (e.g. YEI 1-20 to YEI 1-32) available, with insulation by an oxidized surface. In this case, Table 31 is not applicable.

17.7 Welding

The welding of laminations from C 21 and C 22 depends on the silicon content, the thickness of the insulation and the intensity of grease of the laminations.

The thickness of the insulation and the intensity of grease shall not influence the welding that it is in accordance with the approximate values of Table 32.

Table 32 – Approximate values for the test of welding for laminations from materials C 21 and C 22

Material	Designation	Additional wire	Welding current A	Electrode diameter mm	Electrode distance mm	Welding speed mm/min
C 21	Up to M 330	Yes	70 up to 160	2,4	0,5 to 1	500 up to 700
	Over M 330	No				
C 22	M 165-35S 5	Yes				

NOTE The welding current 70 A is valid for small laminations, the welding current 160 A is valid for large laminations.

Annex A (informative)

Conversion of polarisation and field strength into specific total apparent power

A.1 Specific total apparent power from polarisation and field strength

A direct calculation of the limits given in IEC 60404-8-4 and IEC 60404-8-7 for the magnetising with polarisation $\hat{J}$ and the field strength $\hat{H}$ in a specific total apparent power p_S in VA/kg at a nominal flux density is not practicable because of the non-linearity of the magnetising curve.

For electrical characteristics of laminations a definition of the total apparent power in VA/kg is required.

A conversion of the limits $\hat{J}$ and $\hat{H}$ into an equivalent specific total apparent power is obtained with

- a) a total apparent power under the consideration, that the voltage and the current are sinusoidal at the guarantee values $\hat{J}$ and $\hat{H}$;
- b) the r.m.s. value of the total apparent power at the warranted flux density;
- c) a curve of the specific total apparent power depending on the flux density.

For a): The cardinal equations of the magnetic circuit are:

$$\text{For the voltage:} \quad U = 2 \cdot \pi \cdot f \cdot \frac{\hat{B}}{\sqrt{2}} \cdot A_{Fe} \cdot N \cdot 10^{-4} \quad (\text{A.1})$$

$$\text{For the current:} \quad I = \frac{\hat{H}}{\sqrt{2}} \cdot l_{Fe} \cdot \frac{1}{N} \cdot 10^{-2} \quad (\text{A.2})$$

$$\text{For the core mass:} \quad m_{Fe} = A_{Fe} \cdot l_{Fe} \cdot \rho \cdot 10^{-3} \quad (\text{A.3})$$

From these equations the specific total apparent power $p_{S,\sin}$ is calculated:

$$p_{S,\sin} = \frac{U \cdot I}{m_{Fe}} \quad (\text{A.4})$$

This results in the equation:

$$p_{S,\sin} = \frac{0,157 \cdot \hat{B} \cdot \hat{H}}{\rho} \text{ in VA/kg} \quad (\text{A.5})$$

where

A_{Fe} is the core cross section, in cm²;

$\hat{B}$ is the flux density, in T;

$\hat{H}$ is the field strength, in A/m;

l_{Fe} is the magnetic path length, in cm;

N is the number of turns;

The guarantee values $\hat{H}$ and $\hat{J}$, the density ρ , and the calculated total apparent power $p_{S,\sin}$ for the materials shown in Table 1 are listed in Table A.1.

This specific total apparent power $p_{S,\sin}$ is the reference point resulting from the guarantee values $\hat{J}$ and $\hat{H}$ for the materials.

For materials in accordance with IEC 60404-8-4, the guarantee value at $\hat{H} = 2\,500\text{ A/m}$ is used.

For materials in accordance with IEC 60404-8-7, the field strength is $\hat{H} = 800\text{ A/m}$. In the above-mentioned IEC publications, the polarisation $\hat{J}$ is considered to be identical with the flux density $\hat{B}$. The difference caused by the idle flux density $\mu_0 \cdot \hat{H}$ is negligible. The difference is at $\hat{H} = 2\,500\text{ A/m}$:

$$\mu_0 \cdot \hat{H} = 4 \pi \cdot 10^{-7} \cdot \frac{\text{V} \cdot \text{s}}{\text{A} \cdot \text{m}} \cdot 2500 \cdot \frac{\text{A}}{\text{m}} = 0,00314 \cdot \frac{\text{V} \cdot \text{s}}{\text{m}^2} = 0,00314\text{ T} \quad (\text{A.6})$$

For b): Because of the non-linearity of the magnetic curve, the measured specific apparent power at the guarantee flux density is only 50 % to 60 % of the calculated $p_{S,\sin}$. To eliminate the dispersion of different charges, a total apparent power with 70 % of the calculated $p_{S,\sin}$ at the guarantee flux density is chosen. See p_S at $\hat{B}$ in Table A.1.

For c): The characteristics of the total apparent power depending on the flux density is defined by a mathematical function. With this function, it is possible to convert the guarantee flux density $\hat{B}$ into the requested nominal flux density $\hat{B}_n$ depending on the cardinal value p_{S0} .

The equation for the total apparent power is:

$$p_S \approx p_{BFe} = c_1 \cdot f \cdot \hat{B}^{(2+y)} + c_2 \cdot f^2 \cdot \hat{B}^{(2+y)} \quad (\text{A.7})$$

The characteristics for selected values of the specific reactive power of the core p_{BFe} are shown in Figure A.1. Characteristics for other specific reactive power of the core see Table A.4.

For the characteristics above the guarantee value p_S at the guarantee flux density $\hat{B}$ is determined. Table A.1 shows the data for the C 21 and C 22 materials used in this specification. For power loss in W/kg, see Table 28.

Table A.1 – Data for the specific total apparent power and the specific power loss for C 21 and C 22 materials

Material designation	$\hat{H}$ A/m	$\hat{B}, \hat{J}$ T	At $\hat{B}$		$\hat{B}_n = 1,5 \text{ T}$		$\hat{B}_n = 1,7 \text{ T}$		ρ kg/dm ³	η
			$p_{S,\sin}$ VA/kg	p_S VA/kg	p_{S0} VA/kg	p_{Fe0} W/kg	p_{S0} VA/kg	p_{Fe0} W/kg		
M 270-35A 5	2 500	1,49	76,4	53	56	3	-	-	7,65	0,94
M 330-35A 5						3,75				
M 330-50A 5										
M 400-50A 5		1,51	77,5	53	50	4,5			7,7	0,95
M 530-50A 5		1,56	79,5	55,7	40	6				
M 800-50A 5		1,6	80,5	56,4	31,5	9			7,8	
M 165-35S 5	800	1,75	28,7	21,3	-	-	16	1,8	7,65	0,95

Laminations made from grain-oriented silicon steel have essential higher total apparent power. The cardinal value $p_{S0} = 56 \text{ VA/kg}$ at $\hat{B}_n = 1,7 \text{ T}$ for the laminations is chosen according to the practice. For the grain-oriented materials C 22 only for the magnetising in the rolling direction a guarantee value $\hat{J} = 1,75 \text{ T}$ at $\hat{H} = 800 \text{ A/m}$ is specified. With the same method, a cardinal value $p_{S0} = 16 \text{ VA/kg}$ at $\hat{B}_n = 1,7 \text{ T}$ is found.

The influences of punching, and shape and size of laminations are taken into consideration for the values of specific total apparent power (see Table 29) and for the specific power loss (see Table 30).

The influence on yokes with laminations magnetized perpendicular to the rolled direction is particularly high. The usual shapes of laminations as shown in 13.3 up to 13.10 have areas where the flux direction is mainly in parallel to the rolled direction and other areas where the flux is mainly 90° cross to the rolled direction. For more details see Clause A.7.

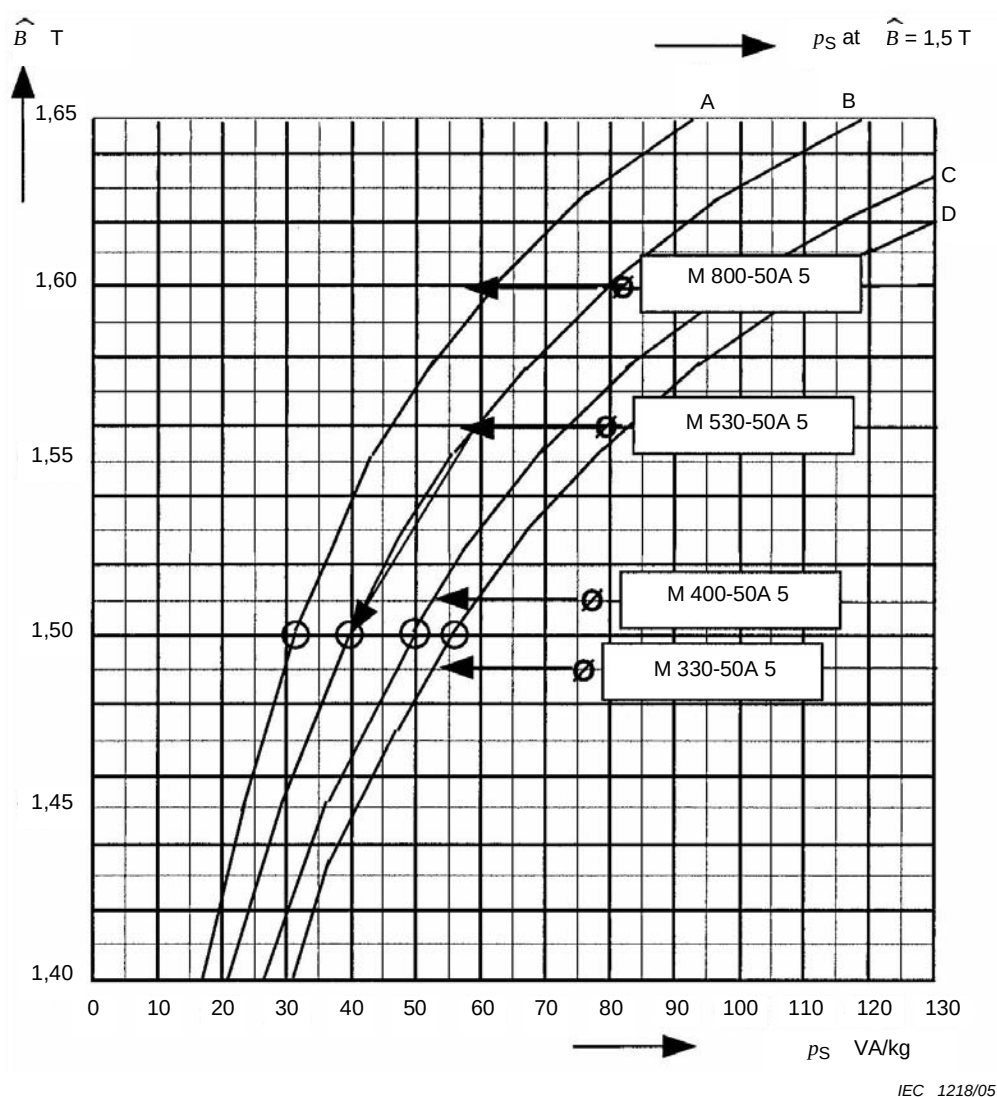
Figure A.1 shows the conversion of sinusoidal total apparent power $p_{S,\sin}$ calculated from guarantee values $\hat{J}$ and $\hat{H}$ into the cardinal value p_{S0} at nominal flux density $\hat{B}_n$.

The characteristics A to D for the total apparent power are taken from Table A.4.

Method:

Example: Material M 530-50A 5 with guaranteed flux density $\hat{B} = 1,56 \text{ T}$.

- 1) Calculate the theoretical sinusoidal total apparent power $p_{S,\sin}$ (see Clause A.5) (in this case $p_{S,\sin} = 79,5 \text{ VA/kg}$);
- 2) Mark this value in Figure A.1 at guaranteed flux density $\hat{B} = 1,56 \text{ T}$ with $\emptyset$;
- 3) Calculate the limit for $p_S = 70 \%$ of $p_{S,\sin}$ (in this case $p_S = 55,7 \text{ VA/kg}$);
- 4) Mark this value at guaranteed flux density $\hat{B} = 1,56 \text{ T}$ with $\leftarrow$;
- 5) Choose the next characteristic on the right position (in this case the characteristic B);
- 6) Follow the characteristic to nominal flux density $\hat{B} = 1,5 \text{ T}$ and mark it with O (in this case the cardinal value $p_{S0} = 40 \text{ VA/kg}$ is corresponding to the cardinal value $p_{Fe0} = 5,3 \text{ W/kg}$).



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Ø: Guarantee value $p_{S,\sin}$

O: Cardinal value p_{S0} for the material

The guarantee value for M 330-50A 5 is also valid for M 330-35A 5 and M 270-35A 5.

Characteristic A: $p_{S0} = 31,5$ VA/kg

Characteristic B: $p_{S0} = 40,0$ VA/kg

Characteristic C: $p_{S0} = 50,0$ VA/kg

Characteristic D: $p_{S0} = 56,0$ VA/kg

Figure A.1 – Specific total apparent power for non-oriented silicon steel C 21

As the specific power losses p_{Fe} are always smaller than the specific reactive power p_B (about $< 20\%$ of p_B), the limit for the specific reactive power is used as the limit of the specific total apparent power.

A.2 Equations for the curves of the magnetic materials

For the calculation of transformers not only one guaranteed point but the curves are required for the specific reactive power and for the specific power loss, depending on the flux density.

For the progress of the curves, equations were developed. The curves will be calculated with constants which are depending on the material and the frequency. The equations and the constants are investigated taking into account a lot of tests and are adjusted to the wanted nominal values at the nominal flux density.

The nominal values at the nominal flux density are in accordance with the R 40 series.

A.2.1 Specific reactive power

The specific reactive power p_B for a transformer consists of the reactive power p_{BFe} of the core and if present the reactive power p_{BL} of the air gap:

$$p_B = p_{BFe} + p_{BL} \text{ in VA/kg} \quad (\text{A.8})$$

A.2.2 Specific reactive power of the material

For the specific reactive power of the material, the equation is:

$$p_{BFe} = c_1 \cdot f \cdot \hat{B}^{(2+y)} + c_2 \cdot f^2 \cdot \hat{B}^{(2+y)} \text{ in VA/kg} \quad (\text{A.9})$$

where $y = 0$ for $\hat{B} \leq \hat{B}_0$ with $\hat{B}$ and $\hat{B}_0$ in dT (deci Tesla)

$$y = y_1 \cdot (\hat{B} - \hat{B}_0)^{y_2} \text{ for } \hat{B} > \hat{B}_0 \quad (\text{A.10})$$

The constants c_1 , c_2 , y_1 , y_2 and $\hat{B}_0$ are depending on the material. In the equations for p_{BFe} and p_{Fe} for the flux density, the unit dT is chosen.

A.2.3 Specific reactive power of the air gap

In cores with an air gap, the specific reactive power is:

$$p_{BL} = 0,25 \frac{l_L \cdot f \cdot \hat{B}^2}{l_{Fe} \cdot \rho} \text{ in VA/kg} \quad (\text{A.11})$$

where

p_{BL} is the specific reactive power of the air gap, in VA/kg;

$\hat{B}$ is the flux density, in T;

l_{Fe} is the magnetic path length, in cm;

l_L is the length of the air gap, in μm ;

ρ is the density of the core, in kg/dm^3 .

A.2.4 Specific power loss

For the specific power loss the equation is:

$$p_{\text{Fe}} = c_3 \cdot f \cdot \hat{B}^{y_3} + c_4 \cdot f^2 \cdot \hat{B}^2 \text{ in W/kg} \quad (\text{A.12})$$

where $\hat{B}$ is the flux density in dT.

The constants c_3 , c_4 , and y_3 are dependent on the material.

A.2.5 Specific total apparent power

The specific total apparent power in a transformer consists of the reactive power and the power loss:

$$p_S = \sqrt{p_B^2 + p_{\text{Fe}}^2} \text{ in VA/kg} \quad (\text{A.13})$$

As the specific power loss p_{Fe} are always smaller than the specific reactive power p_B (about < 20 % of p_B), the limit for the specific reactive power is used as the limit of the specific total apparent power.

A.3 Constants for non-oriented silicon steel C 21

The nominal values for the specific reactive power and the specific power loss are valid for the nominal flux density $\hat{B} = 1,5 \text{ T}$. The constants are valid for the flux density from 1 T to 1,7 T and the frequency from 30 Hz to 700 Hz.

Table A.2 – Constants for the specific reactive power of non-oriented silicon steel C 21

Nominal value	Constants				
p_{BFe} VA/kg	$\hat{B}_0$ dT	c_1	c_2	y_1	y_2
31,5	6	4,47	5,06	$3,37 \times 10^{-3}$	2,4
33,5		4,76	5,22		
35,5		5,04	5,75		
37,5		5,33	5,80		
40		5,68	6,37		
42,5		6,04	6,59		
45		6,39	7,16		
47,5		$6,75 \times 10^{-4}$	$7,38 \times 10^{-7}$		
50		7,10	7,96		
53		7,53	8,26		
56		7,96	8,90		
60		8,52	9,45		
63		8,95	10,10		
67		9,51	10,63		
71		10,09	11,30		
NOTE The nominal values p_{BFe} are valid for $\hat{B} = 1,5 \text{ T}$, $f = 50 \text{ Hz}$.					

**Table A.3 – Constants for the specific power loss
of non-oriented silicon steel C 21.**

Nominal value	Constants		
p_{Fe} W/kg	c_3	c_4	γ_3
3	4,48	7,3	1,75
3,15	4,13	7,9	1,80
3,35	3,84	8,4	1,85
3,55	3,55	9,0	1,90
3,75	3,27	9,6	1,95
4	3,05	10,2	2,00
4,25	2,83	10,8	2,05
4,5	2,61	11,5	2,10
4,75	2,41	12,1	2,15
5	2,22	12,9	2,20
5,3	2,05	13,5	2,25
5,6	1,89	14,3	2,30
6	1,78	14,8	2,35
6,3	1,63	15,8	2,40
6,7	1,51	17,0	2,45
7,1	1,40	17,5	2,50
7,5	1,30	18,0	2,55
8	1,21	19,2	2,60
8,5	1,13	19,7	2,65
9	1,05	20,3	2,70
9,5	0,97	21,0	2,75
10	0,89	22,4	2,80
10,6	0,827	23,2	2,85
NOTE The nominal values p_{Fe} are valid for $\hat{B} = 1,5 \text{ T}$, $f = 50 \text{ Hz}$.			

A.4 Specific reactive power and specific power loss for non-oriented silicon steel C 21

A.4.1 Specific reactive power

Specific reactive power, calculated using equation (A.9) and the constants of Table A.2.

Table A.4 – Specific reactive power for non-oriented silicon steel C 21.

Nominal value p_{BFe} VA/kg	Specific reactive power p_{BFe} in VA/kg									
	at $\hat{B}$ in T									
	1	1,1	1,2	1,3	1,4	1,5	1,55	1,6	1,65	1,7
31,5	2,93	4,20	6,30	10,00	17,1	31,5	44,1	63,2	92,6	139
33,5	3,12	4,62	6,70	10,67	18,2	33,5	46,9	67,2	98,5	148
35,5	3,31	4,73	7,11	11,30	19,3	35,5	49,8	72,3	104,4	157
37,5	3,49	4,99	7,50	11,94	20,4	37,5	52,5	75,2	110,2	166
40	3,73	5,33	8,01	12,70	21,7	40,0	56,1	80,3	118,0	177
42,5	3,95	5,66	8,50	13,54	23,1	42,5	59,5	85,2	125,0	188
45	4,19	6,00	9,01	14,30	24,5	45,0	63,1	90,3	132,0	199
47,5	4,42	6,33	9,50	15,10	25,8	47,5	66,5	95,3	140,0	210
50	4,65	6,66	10,00	15,90	27,2	50,0	70,1	100,0	147,0	221
53	4,93	7,06	10,60	16,90	28,8	53,0	74,2	106,3	156,0	234
56	5,22	7,47	11,20	17,90	30,5	56,0	78,5	112,0	165,0	248
60	5,58	7,99	12,00	19,10	32,6	60,0	84,0	120,0	176,0	265
63	5,87	8,40	12,60	20,10	34,3	63,0	88,4	127,0	185,0	280
67	6,23	8,92	13,40	21,30	36,4	67,0	93,8	134,0	197,0	296
71	6,61	9,47	14,20	22,60	38,6	71,0	100,0	143,0	209,0	314
NOTE The nominal values p_{BFe} are valid for $\hat{B} = 1,5 \text{ T}$, $f = 50 \text{ Hz}$.										

A.4.2 Specific power loss

Specific power loss, calculated using equation (A.12) and constants of Table A.3.

Table A.5 – Specific power loss for non-oriented silicon steel C 21

Nominal value p_{Fe} W/kg	Specific power loss p_{Fe} in W/kg							
	at $\hat{B}$ in T							
	1	1,1	1,2	1,3	1,4	1,5	1,6	1,7
3	1,46	1,72	2,01	2,32	2,65	3,00	3,37	3,75
3,15	1,50	1,79	2,09	2,42	2,77	3,15	3,54	3,96
3,35	1,57	1,88	2,21	2,56	2,95	3,35	3,78	4,23
3,55	1,63	1,96	2,32	2,70	3,11	3,55	4,02	4,51
3,75	1,70	2,05	2,43	2,84	3,28	3,75	4,25	4,79
4	1,78	2,15	2,56	3,00	3,49	4,00	4,56	5,14
4,25	1,86	2,26	2,70	3,17	3,69	4,25	4,85	5,49
4,5	1,93	2,35	2,82	3,34	3,89	4,50	5,14	5,84
4,75	2,00	2,46	2,95	3,50	4,10	4,75	5,45	6,20
5	2,08	2,56	3,09	3,68	4,32	5,00	5,77	6,59
5,3	2,16	2,67	3,23	3,86	4,55	5,30	6,11	6,99
5,6	2,24	2,78	3,38	4,05	4,79	5,60	6,47	7,42
6	2,36	2,94	3,59	4,32	5,12	6,00	6,96	7,88
6,3	2,44	3,05	3,74	4,51	5,37	6,3	7,34	8,46
6,7	2,55	3,20	3,94	4,77	5,69	6,70	7,82	9,04
7,1	2,65	3,34	4,12	5,00	5,99	7,10	8,29	9,61
7,5	2,76	3,49	4,32	5,26	6,32	7,50	8,80	10,20
8	2,89	3,67	4,56	5,58	6,72	8,00	9,40	11,00
8,5	3,02	3,84	4,80	5,89	7,12	8,50	10,00	11,70
9	3,14	4,02	5,04	6,20	7,52	9,00	10,70	12,50
9,5	3,25	4,18	5,26	6,50	7,91	9,50	11,30	13,30
10	3,37	4,34	5,48	6,80	8,30	10,00	11,90	14,00
10,6	3,51	4,54	5,76	7,16	8,77	10,60	12,70	15,00
NOTE The nominal values p_{Fe} are valid for $\hat{B} = 1,5$ T, $f = 50$ Hz.								

A.5 Constants for grain-oriented silicon steel C 22

The nominal values for the specific reactive power and the specific power loss are valid for the nominal flux density $\hat{B} = 1,7$ T. The constants are valid for the flux density for strips from 1 T to 2 T, for laminations from 1 T to 1,8 T and for the frequency from 30 Hz to 700 Hz.

Table A.6 – Constants for the specific reactive power of grain-oriented silicon steel C 22.

Nominal value	Constants					
p_{BFe} VA/kg	$\hat{B}_0$ dT	c_1	c_2	y_1	y_2	Notes
15	11	3,00	5,14	$9,5 \times 10^{-3}$	2,1	Preferred use for strip-wound cores and strip-wound cut cores
16		3,20	5,49			
17		3,40	5,83			
18		$3,60 \times 10^{-4}$	$6,17 \times 10^{-7}$			
19		3,80	6,51			
20		4,00	6,86			
40	6	3,66	6,37	$11,9 \times 10^{-3}$	1,69	Preferred use for contour outline shapes
42,5		3,89	6,74			
45		4,12	7,11			
47,5		4,35	7,49			
50		4,58	7,96			
53		4,85	8,43			
56		5,12	8,99			
60		5,49	9,55			
63		$5,76 \times 10^{-4}$	$10,10 \times 10^{-7}$			
67		6,13	10,68			
71		6,50	11,20			
75		6,86	11,99			
80		7,32	12,70			
85		7,78	13,49			
90		8,24	14,20			
95		8,69	15,18			
100		9,15	15,90			
NOTE The nominal values p_{BFe} are valid for $\hat{B} = 1,7 \text{ T}$, $f = 50 \text{ Hz}$.						

**Table A.7 – Constants for the specific power loss
of grain-oriented silicon steel C 22**

Nominal value	Constants			Notes
p_{Fe} W/kg	c_3	c_4	γ_3	
1,7	0,891	5,7	2,00	Preferred use for strip-wound cores and strip-wound cut cores
1,8	0,869	6,0	2,03	
1,9	0,844	6,3	2,06	
2	0,813	6,7	2,09	
2,12	0,792	7,1	2,12	
2,24	0,768	7,5	2,15	
2,36	0,744	7,9	2,18	Preferred use for contour outline shapes
2,5	0,725	8,3	2,21	
2,65	0,706	8,8	2,24	
2,8	0,685	9,3	2,27	
3	0,674	10,0	2,30	
3,15	0,649	10,5	2,33	
3,35	0,634	11,2	2,36	
3,55	0,618	11,8	2,39	
NOTE The nominal values p_{Fe} are valid for $\hat{B} = 1,7$ T, $f = 50$ Hz.				

A.6 Specific reactive power and specific power loss for grain-oriented silicon steel C 22

A.6.1 Specific reactive power

Specific reactive power, calculated using equation (A.9) and the constants of Table A.6.

Table A.8 – Specific reactive power for grain-oriented silicon steel C 22.

Nominal value p_{BFe} VA/kg	Specific reactive power p_{BFe} in VA/kg at $\hat{B}$ in T							
	1,3	1,4	1,5	1,6	1,7	1,8	1,9	2
17	3,46	4,65	6,66	10,24	17	30,7	60,4	130
18	3,66	4,93	7,06	10,84	18	32,5	63,9	138
19	3,87	5,20	7,45	11,45	19	34,3	67,5	146
20	4,07	5,48	7,84	12,05	20	36,1	71,9	153
40	7,62	11,2	16,8	25,6	40	63,8		
42,5	8,10	11,9	17,8	27,2	42,5	67,8		
45	8,57	12,6	18,9	28,8	45	71,8		
47,5	9,05	13,3	19,9	30,4	47,5	75,8		
50	9,53	14,0	21,0	32,1	50	79,8		
53	10,10	14,8	22,2	34,0	53	84,6		
56	10,7	15,7	23,5	35,9	56	89,3		
60	11,4	16,8	25,2	38,4	60	95,7		
63	12,0	17,6	26,4	40,4	63	100,0		
67	12,8	18,8	28,1	42,9	67	107,0		
71	13,5	19,9	29,8	45,5	71	113,0		
75	14,3	21,0	31,4	48,0	75	120,0		
80	15,2	22,4	33,5	51,2	80	128,0		
85	16,2	23,8	35,6	54,5	85	136,0		
90	17,1	25,2	37,7	57,7	90	144,0		
95	18,1	26,6	39,8	60,9	95	152,0		
100	19,0	28,0	41,9	64,1	100	160,0		
NOTE The nominal values p_{BFe} are valid for $\hat{B} = 1,7 \text{ T}$, $f = 50 \text{ Hz}$.								

A.6.2 Specific power loss

Specific power loss, calculated using equation (A.12) and the constants of Table A.7.

Table A.9 – Specific power loss for grain-oriented silicon steel C22.

Nominal value p_{Fe} W/kg	Specific power loss p_{Fe} in W/kg							
	at $\hat{B}$ in T							
	1,3	1,4	1,5	1,6	1,7	1,8	1,9	2
1,7	0,994	1,15	1,32	1,51	1,7	1,91	2,12	2,35
1,8	1,05	1,22	1,4	1,59	1,8	2,02	2,25	2,5
1,9	1,1	1,28	1,47	1,68	1,9	2,14	2,39	2,65
2	1,15	1,34	1,54	1,76	2	2,25	2,52	2,8
2,12	1,21	1,41	1,63	1,87	2,12	2,39	2,68	2,98
2,24	1,27	1,49	1,72	1,97	2,24	2,53	2,83	3,16
2,36	1,33	1,56	1,81	2,07	2,36	2,67	2,99	3,34
2,5	1,4	1,64	1,91	2,19	2,5	2,83	3,18	3,55
2,65	1,48	1,73	2,02	2,32	2,65	3,0	3,38	3,78
2,8	1,55	1,82	2,12	2,45	2,8	3,18	3,58	4,01
3	1,65	1,95	2,27	2,62	3	3,41	3,85	4,31
3,15	1,72	2,03	2,38	2,75	3,15	3,58	4,04	4,54
3,35	1,82	2,16	2,52	2,92	3,35	3,81	4,31	4,85
3,55	1,92	2,27	2,66	3,09	3,55	4,05	4,58	5,16
NOTE The nominal values p_{Fe} are valid for $\hat{B} = 1,7$ T, $f = 50$ Hz.								

A.7 Influence of the shape on the electrical characteristics for grain-oriented silicon steel C 22

In laminations made of grain-oriented silicon steel from IEC 60404-8-7, there is an influence from the shapes on the total apparent power and the power loss.

The reason is that different parts l_q – length of the yokes – are magnetised perpendicular to the rolling direction, and some with rolling direction.

If the laminations are interleaved throughout the stack one by one in shapes with one yoke in rolling direction and one yoke perpendicular to the rolling direction, a part of the magnetic flux is going into the next lamination.

In shapes with both yokes perpendicular to the rolling direction, the increase of total apparent power and power loss is particularly high.

In Table A.10 for the laminations from 13.3 to 13.10 the parts l_w – length of the part magnetized in rolling direction – to the magnetic path length l_{Fe} and the parts l_q to l_{Fe} are listed.

These influences are considered in the Table 29 for the specific total apparent power and in Table 30 for the specific power loss.

Table A.10 – Parts of the magnetic path in rolling direction l_w / l_{Fe} and perpendicular to the rolling direction l_q / l_{Fe} for laminations of 13.3 to 13.10

Shape type	Part l_w / l_{Fe}	Part l_q / l_{Fe}	Number of yokes perpendicular
YEI 1	0,83	0,17	1
YUI 1	0,83	0,17	1
YUI 2	0,74	0,26	1
YEE 2	0,71	0,29	2
YEI 2	0,85	0,15	1
YEE 2-..L	0,78	0,22	2
YEx 3	0,67	0,33	2
YEI 3	0,83	0,17	1
YEx 4	0,62	0,38	2
YM 1	0,71	0,29	2

Bibliography

IEC 60740-2:1993, *Laminations for transformers and inductors for use in telecommunication and electronic equipment – Part 2: Specification for the minimum permeabilities of laminations made of soft magnetic metallic materials*

IEC 61797-1:1996, *Transformers and inductors for use in telecommunication and electronic equipment – Main dimensions of coil formers – Part 1: Coil formers for laminated cores*



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